Transfer Learning for Meta-analysis Under Covariate Shift

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***Abstract*—Randomized controlled trials often do not represent the populations where decisions are made, and covariate shift across studies can invalidate standard IPD meta-analysis and transport estimators. We propose a placebo-anchored transport framework that treats source-trial outcomes as abundant proxy signals and target-trial placebo outcomes as scarce, high-fidelity gold labels to calibrate baseline risk. A low-complexity (sparse) correction anchors proxy outcome models to the target population, and the anchored models are embedded in a cross-fitted doubly robust learner, yielding orthogonal estimation of patient-level heterogeneous treatment effects when target treated outcomes are available. We distinguish two regimes: in connected targets (with a treated arm), the method yields target-identified effect estimates; in disconnected targets (placebo-only), it reduces to a principled screen–then–transport procedure under explicit working-model transport assumptions. Experiments on synthetic data and a semi-synthetic IHDP benchmark evaluate point-wise CATE accuracy, ATE error, ranking quality for targeting, decision-theoretic policy regret, and calibration. Across connected settings, the proposed method is best or near-best and improves substantially over proxy-only, target-only, and transport baselines at small target sample sizes; in disconnected settings, it retains strong ranking performance for targeting while pointwise accuracy depends on the strength of the working transport condition.**

***Index Terms*—Transfer learning, meta-analysis, causal inference, clinical trial, doubly robust estimation.**

I. Introduction

Randomized controlled trials (RCTs) remain the gold standard for estimating treatment effects [1], and drug approvals [2]. Yet, their external validity or generalizability remains a major challenge,as trial participants often differ from the patient population of interest [3], thus decision-makers often need to know how a therapy would perform in a different target population than the one studied. This challenge arises frequently in oncology and other therapeutic areas where multiple trials are conducted in distinct populations, and where no single trial provides a direct comparison in the population of interest.

Network meta-analysis (NMA) has become the dominant framework for combining this evidence, allowing estimation of relative treatment effects across a connected network of trials [4, 5]. When individual patient data (IPD) are available, IPD-NMA can incorporate treatment–covariate interactions to explore the heterogeneity of the effect and, in principle, estimate treatment effects in specific subpopulations [6]. However, standard IPD-NMA methods rely on two key conditions: Network connectivity – the target trial must be connected to the network through a shared comparator arm (e.g., placebo or standard of care). Shared-comparator comparability – the comparator arms across trials must be exchangeable after adjustment for measured. [7].

In practice, these conditions are often violated. Network connectivity is violated if target trials are disconnected, for example containing only a placebo arm for the treatment of interest. Baseline risks in shared comparators differ across studies due to covariate shift and unmeasured effect modifiers, violating shared-comparator comparability.

IPD-NMA typically models relative effects without anchoring to the observed absolute risk in the target population, leaving results vulnerable to bias when proportional hazards do not hold or when hazard ratios exhibit non-collapsibility. Furthermore, standard NMA outputs are network-wide averages; transporting them to a specific target cohort usually requires meta-regression or marginalization, not patient-level counterfactual prediction.

These limitations create a methodological gap: there is no established approach that can (i) transport trial evidence to a target population lacking a concurrent treated arm, (ii) correct for covariate shift in both measured covariates and residual baseline risk, and (iii) support patient-level, target-specific decision analysis in disconnected trial settings.

We propose a placebo-anchored transport framework to address this gap. Our method learns conditional outcome models from source-trial IPD (treatment and placebo), explicitly calibrates baseline risk using observed placebo outcomes in the target trial, and embeds the calibrated models in an orthogonal estimation pipeline. This approach requires no network connectivity and directly adjusts for covariate shift through anchoring. When treated outcomes are observed in the target (even if limited), it yields individualized counterfactual predictions via a doubly robust learner; when treated outcomes are unavailable, it provides calibrated baseline risk

Code available at: https://github.com/wangzilongri/SparseTLDRICHI2026 and a principled working-model extrapolation for scenario and sensitivity analysis.

II. Related Work

IPD meta-analysis and transportability methods provide the main alternatives for combining trial evidence. Standard IPD-NMA [6, 4] requires network connectivity and shared-comparator comparability; when the target is disconnected (e.g., placebo-only) or under covariate shift, these conditions fail. Transportability and weighting approaches [8, 9] formalize external validity via selection diagrams and reweighting (IPW [10, 11], AIPW [12], entropy balancing [13]), but rely on overlap, correct propensity specification, and do not anchor to target placebo outcomes. None of these directly address calibration of proxy outcome models using scarce target placebo data in disconnected settings.

- **IPD-NMA and transportability:** Network meta-analysis and transport estimators [6, 8] assume connectivity or sufficient overlap; they do not use placebo-at-target as a gold calibration signal.
- **Proxy–gold and transfer learning:** High-dimensional transfer learning [14, 15] treats abundant source data as proxy and scarce target data as gold; we adopt this view and anchor to target placebo.
- **Orthogonal and DR learners:** Doubly robust and DML estimators [16, 17] give Neyman-orthogonal CATE estimation; we combine placebo-anchored outcome models with cross-fitted DR learning.
- **Other approaches:** Classical parametric IPD meta-analysis, ML (e.g., causal forests [18], BART[19]), and TMLE [20] improve flexibility but typically assume connectivity or lack explicit placebo anchoring; deep causal methods exist but share similar limitations.

We therefore propose a placebo-anchored transport framework that calibrates proxy models to target placebo outcomes and embeds them in an orthogonal DR pipeline, without requiring network connectivity.

A. Our Contributions

This paper develops a calibrated framework for transporting individualized treatment effect estimates across heterogeneous randomized controlled trials under covariate shift. We formulate the transport problem as a *proxy–gold* learning task in the sense of high-dimensional transfer learning, where outcomes from source trials provide abundant but potentially miscalibrated proxy signals, and placebo outcomes observed at the target site provide a limited but informative calibration signal for baseline risk. This formulation does not claim causal identification of target treatment effects from target data alone; instead, it defines a stable working-model estimand suitable for disconnected or weakly connected trial settings.

Within this framework, we introduce a placebo-anchored correction mechanism that adjusts proxy outcome regressions using a restricted, low-complexity function class. The resulting anchoring step explicitly targets systematic baseline miscalibration induced by population heterogeneity and yields calibrated outcome predictions in the target population. The low-complexity restriction is imposed as a regularity condition rather than an identification assumption, allowing estimation error to be decomposed into stochastic error and an explicit structural approximation term.

We further embed the anchoring procedure in a doubly robust learner with cross-fitting, exploiting the fact that treatment assignment is randomized and propensities are known by design in randomized trials. The resulting estimator is Neyman-orthogonal and admits a $\sqrt{N}$ asymptotic linear expansion for the working-model conditional average treatment effect (CATE), where N denotes the total sample size across sites. The analysis clarifies that while the leading stochastic term is governed by N, finite-sample performance depends jointly on the proxy sample size, the number of target placebo observations used for calibration, and approximation error arising from violations of the working model. Proofs and detailed rate and error statements appear in the appendix (Section A).

Finally, through synthetic and semi-synthetic experiments, including ablations and robustness checks, we empirically characterize the operating regimes of placebo anchoring and illustrate how performance degrades gracefully toward proxy-only baselines as structural assumptions are weakened.

B. Organization of this Paper

The remainder of the paper proceeds as follows. § III formalizes the problem setting, data structure, and identification assumptions. § IV presents the placebo-anchored proxy–gold estimator and the DR-Learner integration in detail. § V evaluates the approach on synthetic ablations, along with robustness checks when some identification assumptions are violated. § VI evaluates the approach on a semi-synthetic benchmark derived from the IHDP dataset. Finally, § VII summarizes findings, discusses limitations, and outlines directions for future work.

III. Preliminaries

Here we introduce our preliminaries and notation for this paper.

A. Problem Statement

We study the setting where a therapy is evaluated across multiple RCTs, each conducted at a different site or in a different patient population. Within every site, both treatment and placebo arms are observed under a harmonized protocol (identical dosage, randomization scheme, and eligibility criteria). Consequently, treatment assignment is randomized and the propensity model is known by design.

The methodological challenge arises because the covariate distribution of enrolled patients drifts across sites, reflecting population heterogeneity (e.g., the same oncology drug trialed in the U.S. versus Norway). Such covariate shift induces systematic differences in baseline risk and treatment–covariate interactions, leading to biased estimates if outcomes are naively pooled or if network connectivity assumptions are imposed.

Our goal is to estimate patient-level counterfactual outcomes and treatment effects in a target site by leveraging information from source trials while anchoring estimates to the observed placebo outcomes in the target. Following [14]'s proxy–gold paradigm, we treat source-site outcomes as `proxy` labels and the target-site placebo outcomes as scarce but high-fidelity `gold` labels. This enables calibrated transport of treatment effects across heterogeneous populations without requiring network connectivity.

B. Data and Proxy–Gold Setup

We observe multiple randomized controlled trials (RCTs) indexed by $c \in \{1, 2, \ldots, C\}$ (`sources`) together with one `target` trial, indexed by $c = 0$. In each trial c, let $X \in \mathbb{R}^p$ denote baseline covariates, $A \in \{0, 1\}$ the randomized treatment assignment, and Y the outcome. Because treatment is randomized within each trial, the propensity $e_c(x) = \mathbb{P}(A = 1 \mid X = x, c)$ is known by design.

Proxy data (abundant). From the source trials we have individual participant data (IPD) for both treatment and placebo arms,

$$\mathcal{D}_{\text{proxy}} = \bigcup_{c=1}^{C} \{(X_i^{(c)}, A_i^{(c)}, Y_i^{(c)})\}_{i=1}^{n_c}.$$

These data allow us to fit rich conditional outcome models. However, due to population heterogeneity and covariate shift across sites, such models may be biased when applied directly to the target trial.

Gold data (scarce but high-fidelity). In the target trial $c = 0$, both treatment and placebo arms exist, but the placebo outcomes play a special role: they directly reveal the target population's baseline risk distribution. We denote

$$\mathcal{D}_{\text{gold}} = \{(X_j^{(0)}, A_j^{(0)} = 0, Y_j^{(0)})\}_{j=1}^{m}.$$

$\mathcal{D}_{\text{proxy}}$ provides a low-variance but potentially biased signal, whereas $\mathcal{D}_{\text{gold}}$ provides a scarce but unbiased calibration signal. We therefore cast the target placebo outcomes as "gold" labels that anchor and debias outcome models trained on the source trials.

C. Identification and Regularity Assumptions

We distinguish carefully between *design-based identification* and *model-based regularity conditions*. Our approach does *not* assume classical cross-site exchangeability, nor does it claim nonparametric identification of the true target-site treatment effect. Instead, we define and estimate a *working-model CATE* whose interpretation and stability rely on explicit structural regularity conditions.

A1 Consistency. For each individual,

$$Y = Y(1)\,\mathbb{I}(A = 1) + Y(0)\,\mathbb{I}(A = 0).$$

A2 Randomization within sites. Within each trial c, treatment assignment is randomized:

$$(Y(0), Y(1)) \perp A \mid X,\, c.$$

Consequently, the propensity score $e_c(x) = \mathbb{P}(A = 1 \mid X = x, c)$ is known by design.

A3 Positivity (overlap). There exists $\epsilon > 0$ such that for all sites c and all x in the support,

$$\epsilon \leq e_c(x) \leq 1 - \epsilon.$$

This is a design property of randomized controlled trials and applies only to baseline covariates measured prior to treatment assignment.

A4 Outcome-regression stability (regularity). For each arm $a \in \{0, 1\}$ and site c, the conditional outcome regression

$$\mu_{a,c}(x) := \mathbb{E}[Y \mid A = a, X = x, c]$$

is Lipschitz continuous in x with a site-uniform constant $L < \infty$:

$$\left|\mu_{a,c}(x) - \mu_{a,c}(x')\right| \leq L\|x - x'\|, \quad \forall x, x'.$$

This assumption is a *regularity condition* ensuring that moderate covariate shift does not induce arbitrarily large changes in predicted outcomes. It does *not* imply exchangeability or transportability across sites.

A5 Low-complexity transport bias (approximation condition). Differences between site-specific outcome regressions and proxy regressions are structured and of low complexity. Specifically, for each arm a and site c,

$$\mu_{a,c}(x) = \mu_a^{\text{proxy}}(x) + \delta_{a,c}(x),$$

where $\delta_{a,c}$ belongs to a restricted function class $\mathcal{D}$ (e.g., sparse linear functions $\delta(x) = \beta^\top x$ with $\|\beta\|_0 \leq s \ll p$). This assumption is *not* an identification assumption. It defines a working approximation class that enables stable estimation and calibrated extrapolation. When A5 holds approximately, estimation error decomposes into stochastic error and approximation error; when violated, performance degrades smoothly toward proxy-only baselines.

A6 Screening-valid transportability under placebo-based source selection (working-model restriction). In disconnected targets (or whenever treated outcomes in the target are unavailable or not used), our `Proposed-B` procedure selects a subset of source trials using *placebo-arm* compatibility with the target, and then learns a DR CATE model on the selected sources which is transported to the target by prediction. Accordingly, we assume the placebo-based screening step is informative for CATE transportability in the following sense.

There exists a (possibly unknown) subset of sources $\mathcal{C}_0^\star \subseteq \{1, \ldots, C\}$ and constants $(\epsilon_0, \epsilon_\tau) \geq 0$ such that:

(a) **Placebo compatibility on the target support.** For all $c \in \mathcal{C}_0^\star$,

$$\left\|\mu_{0,c} - \mu_{0,0}\right\|_{L_2(P_0)} \leq \epsilon_0,$$

where P_0 denotes the distribution of $X \mid S = 0$ and $\mu_{a,c}(x) = \mathbb{E}[Y \mid A = a, X = x, S = c]$.

(b) **CATE proximity on the target support.** For all $c \in \mathcal{C}_0^\star$,

$$\begin{aligned}\left\|\tau_c-\tau_0\right\|_{L_2(P_0)} &\le \epsilon_\tau \\ \tau_c(x) &:= \mu_{1,c}(x)-\mu_{0,c}(x) \\ \tau_0(x) &:= \mu_{1,0}(x)-\mu_{0,0}(x).\end{aligned}$$

Moreover, the placebo-based detection step applied to the target placebo sample $\mathcal{D}_{0,0} = \{(X_i, Y_i) : S_i = 0, A_i = 0\}$ identifies (up to small error) a subset contained in $\mathcal{C}_0^\star$; i.e., with probability at least $1-\eta$,

$$\widehat{\mathcal{C}}_0 \subseteq \mathcal{C}_0^\star.$$

Interpretation. Assumption **A6** is *not* a design-based identification condition. It defines the *working transport regime* in which placebo-based screening yields a set of sources whose CATE functions are (approximately) transportable to the target on the target covariate support. The approximation level ϵ_τ explicitly captures irreducible cross-arm non-transfer not detectable from placebo outcomes alone; when ϵ_τ is non-negligible, `Proposed-B` targets a stable working-model limit rather than the true τ_0.

Interpretation. Assumptions **A1–A3** are standard design-based conditions for randomized trials. Assumptions **A4–A6** are regularity and approximation conditions that define a well-posed working estimand rather than a nonparametrically identified causal effect. Throughout, we emphasize calibrated estimation under acknowledged transport bias rather than causal identification under strong exchangeability assumptions.

IV. Proposed Framework

A central premise of our framework is that every trial is randomized, so the true propensities $e_c(x)$ are known by design. This ensures that any doubly robust estimator built on top of site-specific outcome models is consistent as long as the outcome regressions are well calibrated. The key methodological challenge is therefore not treatment assignment, but *covariate shift across sites*: outcome models trained on sources may be systematically biased when applied to the target population. We address this challenge by treating source trial outcomes as `proxy` labels and target placebo outcomes as scarce but high-fidelity `gold` labels for calibration. This can be seen as a combination of [14]'s multi-task transfer learner for data-fusion and [21]'s Doubly Robust (DR)-Learner

A. Proposed Estimator

We build on the glmtrans framework [15] for transfer learning under high-dimensional GLMs. Our approach applies glmtrans separately to each treatment arm, yielding target-anchored outcome models that are then combined into CATE estimates. We describe three variants: a plug-in estimator (`Proposed`), a doubly robust variant with cross-fitting (`Proposed-CF`), and an extension for disconnected targets where no treated outcomes are observed (`Proposed-B`).

1) Stage 1: Arm-Specific Transfer Learning via glmtrans: For each treatment arm $a \in \{0,1\}$, we estimate a target-specific outcome regression $\widehat{\mu}_a(x)$ using the two-step glmtrans algorithm with automatic source detection. Let $\mathcal{D}_{a,0} = \{(X_i, Y_i) : c_i = 0, A_i = a\}$ denote the target arm-a data and $\mathcal{D}_{a,c}$ the corresponding source data for site $c \ge 1$.

Source Detection (Trans-GLM). We first identify which sources are statistically compatible with the target using cross-validation. Partition the target data into three folds $\{\mathcal{D}_{a,0}^{[r]}\}_{r=1}^3$. For each source c and each hold-out fold r, compute the cross-validated prediction loss:

- $\widehat{L}_0^{(c)[r]}$: loss on fold r after pooling source c with folds $\{1,2,3\} \setminus \{r\}$
- $\widehat{L}_0^{(0)[r]}$: loss on fold r using only target folds $\{1,2,3\} \setminus \{r\}$ (Lasso baseline)

Average across folds: $\widehat{L}_0^{(c)} = \frac{1}{3}\sum_{r=1}^3 \widehat{L}_0^{(c)[r]}$. Source c is deemed *transferable* if $\widehat{L}_0^{(c)} - \widehat{L}_0^{(0)} \le C_0 \cdot \widehat{\sigma}$, where $\widehat{\sigma}$ is the empirical standard deviation of the baseline losses and $C_0 > 0$ is a threshold constant (default $C_0 = 2$). Let $\widehat{\mathcal{A}}_a \subseteq \{1, \dots, C\}$ denote the selected transferable sources for arm a.

Two-Step Transfer ($\widehat{\mathcal{A}}$-Trans-GLM). Given the selected sources $\widehat{\mathcal{A}}_a$, we apply the two-step transfer algorithm:

1) **Transferring step:** Pool selected sources with target and fit an ℓ_1-penalized GLM:

$$\widehat{w}^{\widehat{\mathcal{A}}_a} \in \arg\min_w \left\{ \sum_{c \in \{0\} \cup \widehat{\mathcal{A}}_a} \sum_{i \in \mathcal{D}_{a,c}} (Y_i - X_i^\top w)^2 + \lambda_w \|w\|_1 \right\}.$$

This pooled estimator has low variance but may be biased for the target due to cross-site heterogeneity.

2) **Debiasing step:** Correct for target-specific bias using only target data:

$$\widehat{\delta}_a \in \arg\min_\delta \left\{ \sum_{i \in \mathcal{D}_{a,0}} \left(Y_i - X_i^\top(\widehat{w}^{\widehat{\mathcal{A}}_a} + \delta)\right)^2 + \lambda_\delta \|\delta\|_1 \right\}.$$

The final target-anchored estimate is $\widehat{\beta}_a = \widehat{w}^{\widehat{\mathcal{A}}_a} + \widehat{\delta}_a$, yielding $\widehat{\mu}_a(x) := x^\top \widehat{\beta}_a$. In our implementation we use the `glmtrans` R package [15] with `glmnet`: covariates are standardized (zero mean, unit variance) before the ℓ_1-penalized fits so the penalty acts uniformly; coefficients are back-transformed to the original scale when reporting. The regularization path is computed on a logarithmic grid (as in `glmnet`), and λ_w, λ_δ are chosen by K-fold cross-validation minimizing mean squared error (`lambda.min`).

Remark. The debiasing step is equivalent to [14]'s proxy–gold correction: the pooled fit $\widehat{w}^{\widehat{\mathcal{A}}_a}$ serves as a variance-reduced "proxy" while the target-only correction $\widehat{\delta}_a$ anchors the estimate to "gold" target data. When no sources pass detection, the algorithm reduces to target-only Lasso, automatically guarding against negative transfer.

2) Stage 2: CATE Estimation: **Option A (`Proposed`): Plug-in CATE.** When the target contains both placebo ($m_0 > 0$) and treated ($m_1 > 0$) observations, we apply glmtrans separately to each arm and compute the plug-in CATE:

$$\widehat{\tau}(x) := \widehat{\mu}_1(x) - \widehat{\mu}_0(x).$$

Option A with DR (`Proposed-CF`): Cross-Fitted Doubly Robust. To reduce sensitivity to outcome model misspecification, we construct doubly robust pseudo-outcomes on the target data. Using K-fold cross-fitting (typically $K = 2$), for each fold k:

1) Fit $\widehat{\mu}_0^{(-k)}, \widehat{\mu}_1^{(-k)}$ on out-of-fold data
2) Compute pseudo-outcomes on fold k:

$$\psi_i := \widehat{\mu}_1^{(-k)}(X_i) - \widehat{\mu}_0^{(-k)}(X_i) + \frac{A_i - e(X_i)}{e(X_i)(1 - e(X_i))}\Big(Y_i - \widehat{\mu}_{A_i}^{(-k)}(X_i)\Big), \tag{1}$$

where $e(X_i)$ is the known propensity score.

The final DR-CATE is obtained by regressing ψ_i on X_i via ℓ_1-penalized regression:

$$\widehat{\tau}_{\mathrm{DR}}(\cdot) \approx \mathbb{E}[\psi \mid X = \cdot].$$

Under standard DML regularity conditions [16], this estimator is $\sqrt{n}$-consistent and robust to first-stage estimation errors. Formal statements and proofs of the convergence rates and error decomposition for both connected and disconnected regimes are given in Appendix A.

Option B (`Proposed-B`): Disconnected Targets. When the target site has no treated outcomes ($m_1 = 0$), we cannot apply glmtrans to the treated arm in the target. Instead, we use target placebo data for source detection and learn the CATE model entirely from selected sources:

1) **Source detection:** Apply Trans-GLM using only target placebo outcomes $\{(X_i, Y_i) : c_i = 0, A_i = 0\}$ to identify transferable sources $\widehat{\mathcal{A}}$.
2) **Source-DR learning:** On the selected source data, fit outcome models $\widehat{\mu}_0^{\mathrm{src}}, \widehat{\mu}_1^{\mathrm{src}}$ via Lasso, compute DR pseudo-outcomes using (1), and fit a CATE model $\widehat{\tau}^{\mathrm{src}}(x)$ on the source pseudo-outcomes.
3) **Transport:** Apply the source-learned CATE directly to the target: $\widehat{\tau}_{\mathrm{target}}(x) := \widehat{\tau}^{\mathrm{src}}(x)$.

This approach assumes the CATE function is transportable from sources to target—a weaker condition than requiring outcome model transportability for both arms separately.

V. Experiments

We evaluate the framework through synthetic ablations and robustness checks, addressing (i) contribution of anchoring and orthogonalization, (ii) scaling with target size, source availability, and dimension, and (iii) degradation under Assumption **A5** violations. We compare against the following baselines:

- `TargetOnly`: Doubly robust learner [17] using only target data (no transfer);
- `ProxyOnly`: Pooled source outcome models applied directly to target (no anchoring);
- `IPW-Transport`, `OM-Transport`: Standard transportability estimators [8];
- `EntropyBal`: Entropy balancing for covariate shift adjustment [13];
- `AnchorOnly`: Placebo-anchored with DR orthogonalization;
- `Proposed`: Our method based on ℓ_1-penalized transfer learning [15] with automatic source selection. Variants include `Proposed-CF` (cross-fitted) and `Proposed-B` (Option B for disconnected targets with $m_1 = 0$).

All experiments use $R = 100$ Monte Carlo replicates per scenario. Ground-truth potential outcomes are known in all synthetic settings, enabling direct evaluation of counterfactual prediction accuracy and CATE ranking quality.

A. Evaluation Metrics

We report PEHE, ATE error, Spearman rank correlation, policy regret, and CATE calibration; full results for all metrics and sweeps are in Appendix C.

CATE accuracy (pointwise). We evaluate individualized treatment effect estimation by the *Precision in Estimating Heterogeneous Effects* (PEHE):

$$\mathrm{PEHE} := \sqrt{\frac{1}{n}\sum_{i=1}^{n}\big(\widehat{\tau}(X_i) - \tau(X_i)\big)^2},$$

where $\tau(x) = \mu_{1,0}(x) - \mu_{0,0}(x)$ is the ground-truth target-site CATE. Lower is better.

ATE accuracy (marginal). We report the absolute ATE error $|\widehat{\mathrm{ATE}} - \mathrm{ATE}|$, where $\mathrm{ATE} = \mathbb{E}[\tau(X)]$ under the target covariate distribution. Lower is better.

Ranking quality. Since clinical decisions are often resource-constrained, we evaluate how well methods rank patients by true uplift using **Spearman** rank correlation between $\widehat{\tau}(X_i)$ and $\tau(X_i)$. Higher is better; values approaching 1 indicate oracle-like targeting.

Policy regret. We evaluate decision relevance using a threshold policy $\widehat{\pi}(x) = \mathbb{I}\{\widehat{\tau}(x) > 0\}$. Define the oracle-evaluable policy value $V(\pi) := \mathbb{E}[\mu_{0,0}(X) + \pi(X)\,\tau(X)]$. We report **policy regret** $V(\pi^\star) - V(\widehat{\pi})$, where $\pi^\star$ is the oracle policy. Lower regret is preferred.

CATE calibration. We assess whether predicted CATEs are well-calibrated by regressing true $\tau(X_i)$ on predicted $\widehat{\tau}(X_i)$. A well-calibrated estimator has **slope** ≈ 1 and **intercept** ≈ 0. We also report calibration $\mathbf{R}^2$ (higher is better) and the **Expected Calibration Error** (ECE), which measures the average absolute difference between predicted and true CATEs within quantile bins. Lower ECE is better.

B. Data-Generating Process (DGP)

We generate multi-center RCT data with explicit covariate shift, controlled signal-to-noise ratios, and known ground-truth treatment effects. The DGP creates fair evaluation conditions where transfer learning can plausibly help, while avoiding adversarial or unrealistically favorable settings.

Covariates and sites. We simulate $C = 10$ source sites and one target site ($c = 0$). For each subject, we sample p-dimensional baseline covariates $X \in \mathbb{R}^p$, with $p \in \{10, 20, 50, 100\}$ varied across experiments. Site-level covariate shift is induced via $X \mid c \sim \mathcal{N}(\mu_c, I)$, calibrated to produce moderate overlap (AUC ≈ 0.75 for source vs. target classification).

TABLE I

AVERAGE RANK PER METRIC ACROSS ALL SWEEPS (1 = BEST). CALIBRATION: SLOPE (IDEAL=1), R^2, ECE (EXPECTED CALIBRATION ERROR).

	Performance				Calibration		
Method	PEHE	ATE	τ	Regret	Slope	R^2	ECE
TargetOnly	6.9	3.3	7.0	6.8	8.3	7.0	6.4
ProxyOnly	8.6	8.2	8.6	8.6	7.0	8.5	8.3
IPW-Transport	4.5	6.6	4.4	4.5	4.5	4.4	5.2
OM-Transport	3.7	5.7	3.5	3.7	3.2	3.4	4.2
EntropyBal	5.3	6.3	5.2	5.3	5.4	5.2	5.7
AnchorOnly	7.7	3.1	7.6	7.4	8.2	7.8	6.9
Proposed	**1.4**	**1.2**	**1.3**	**1.4**	3.5	**1.3**	1.6
Proposed-CF	1.6	1.7	2.1	1.8	**1.9**	2.1	**1.3**
Proposed-B	2.7	5.0	2.8	2.8	2.7	2.7	3.7

Treatment assignment. Treatment is randomized with logistic propensity $e(X)$. Propensity scores are known by design.

Outcomes. Outcomes follow the proxy–deviation decomposition from Assumption **A5**:

$$\mu_{a,c}(x) = \mu_a^{\text{proxy}}(x) + \delta_{a,c}(x),$$

where $\mu_a^{\text{proxy}}(x) = x^\top \beta_a$ are shared outcome regressions and $\delta_{a,c}(x) = x^\top \gamma_{a,c}$ are sparse site-specific deviations. The *nontransfer scale* σ_ν controls the magnitude of non-transferable variation; we use $\sigma_\nu = 0.1$ (SNR $\approx$ 3–4) as the fair baseline.

Target and source budgets. We vary the target budget (m_0, m_1), where m_0 is the placebo sample size and m_1 is the treated sample size. Setting $m_1 = 0$ (placebo-only target) tests Option B methods for disconnected settings. Source sites contribute 20,000 total observations (2,000 per site).

Implementation. All simulations are implemented in Python with fixed random seeds to ensure reproducibility.

C. Summary Across Metrics

Table I reports average rank (1 = best) per metric across all sweeps. `Proposed` leads on PEHE, ATE, Spearman, and Regret; `Proposed-CF` leads on calibration slope and ECE. This trade-off is expected: the plug-in `Proposed` uses the same outcome models for CATE prediction and can achieve lower pointwise error (PEHE), while `Proposed-CF` uses cross-fitting to form doubly robust pseudo-outcomes, which reduces overfitting and improves CATE calibration (slope $\approx$ 1, lower ECE) at the cost of slightly higher PEHE in finite samples. In practice, `Proposed` is preferred when pointwise accuracy is the priority; `Proposed-CF` is preferred when well-calibrated CATEs or inference are needed. Full calibration tables are in Appendix C.

D. Target Budget vs. Dimensionality

Table II shows PEHE across target budgets (m_0, m_1) and dimensions $p \in \{10, 20, 50, 100\}$.

Key findings. (i) `TargetOnly` deteriorates as p grows relative to target size; `Proposed` achieves lowest PEHE when target treated data exist, with `Proposed-CF` helping at higher p. (ii) The $(m_1 = 0)$ column is Option B (disconnected); `Proposed-B` produces valid estimates with strong performance. Full results in Appendix C.

TABLE II

PEHE ACROSS BUDGET (m_0, m_1) AND DIMENSIONALITY p. LOWER IS BETTER.

p	Method	50/0	150/100	550/500
10	TargetOnly	–	1.43±0.39	1.38±0.45
	ProxyOnly	2.34±0.67	1.96±0.66	2.10±0.83
	IPW-Transport	1.80±0.83	1.60±0.70	1.82±1.06
	OM-Transport	1.80±0.83	1.60±0.70	1.81±1.06
	EntropyBal	1.81±0.83	1.60±0.70	1.82±1.06
	AnchorOnly	–	1.46±0.41	1.39±0.46
	Proposed	–	0.43±0.16	**0.28±0.14**
	Proposed-CF	–	**0.41±0.15**	**0.28±0.14**
	Proposed-B	**1.63±0.75**	1.51±0.66	1.81±1.06
20	TargetOnly	–	2.60±0.64	2.33±0.50
	ProxyOnly	3.37±0.84	3.15±0.86	3.02±0.84
	IPW-Transport	2.24±1.10	2.26±1.06	2.21±0.96
	OM-Transport	2.24±1.10	2.27±1.07	2.20±0.96
	EntropyBal	2.25±1.10	2.27±1.06	2.21±0.96
	AnchorOnly	–	2.65±0.71	2.34±0.51
	Proposed	–	0.52±0.13	0.35±0.16
	Proposed-CF	–	**0.50±0.14**	**0.33±0.17**
	Proposed-B	**1.95±0.92**	2.25±1.06	2.14±0.98
50	TargetOnly	–	4.75±0.86	4.57±0.80
	ProxyOnly	5.78±1.22	5.30±1.00	5.29±1.04
	IPW-Transport	3.26±1.83	2.98±1.46	3.17±1.52
	OM-Transport	3.25±1.82	2.95±1.47	3.09±1.53
	EntropyBal	3.40±1.79	3.05±1.44	3.18±1.52
	AnchorOnly	–	4.80±0.90	4.63±0.82
	Proposed	–	**0.73±0.16**	0.46±0.12
	Proposed-CF	–	1.19±0.43	**0.40±0.12**
	Proposed-B	**3.25±1.82**	2.53±1.24	3.03±1.55
100	TargetOnly	–	7.57±1.58	7.08±1.03
	ProxyOnly	8.21±1.23	8.00±1.70	7.73±1.30
	IPW-Transport	4.10±2.05	4.32±2.43	4.61±1.94
	OM-Transport	4.06±2.06	4.21±2.45	4.28±1.98
	EntropyBal	4.77±1.93	4.67±2.39	4.81±1.94
	AnchorOnly	–	7.64±1.67	7.18±1.06
	Proposed	–	**1.71±0.75**	0.58±0.11
	Proposed-CF	–	3.22±1.66	**0.52±0.11**
	Proposed-B	**4.05±2.06**	4.20±2.45	3.96±2.14

E. Source Site Scaling

Table III shows PEHE as the number of source sites $C \in \{2, 5, 10, 20, 50\}$ varies (1,000 samples per site).

Key findings. (i) Adding sources reduces PEHE at small target budgets; (ii) `Proposed` benefits from source diversity via automatic source selection. Full results in Appendix C.

F. Sensitivity to A5 Violations

We vary sparsity $s/p \in \{0.05, 0.5, 1.0\}$ and nonlinearity $\Gamma \in \{0, 0.5, 1\}$ (linear deviation $\delta_{a,c}(x) = (1 - \Gamma)x^\top \gamma_{a,c} + \Gamma g(x)$ with $g(x) = \sum_j \tanh(x_j)$). Table IV shows PEHE.

Key findings. (i) At $(s/p = 0.05, \Gamma = 0)$ (A5 holds), `Proposed` achieves lowest PEHE; (ii) performance degrades smoothly with violations (no catastrophic failure). Full results in Appendix C.

G. Disconnected Regime (Placebo-Only Target)

The disconnected regime $(m_1 = 0)$ is non-identified from target data; Option B is a working-model transport estimator under A6. Table V shows PEHE for $m_0 = 50$,

TABLE III
PEHE ACROSS SOURCE SITES C AND BUDGET. LOWER IS BETTER.

C	Method	50/0	150/100	550/500
2	TargetOnly	–	4.95±1.18	4.46±0.91
	ProxyOnly	5.67±1.05	5.55±1.37	5.19±1.22
	IPW-Transport	3.64±1.45	3.78±1.85	3.74±1.97
	OM-Transport	3.54±1.45	3.59±1.89	3.52±1.98
	EntropyBal	3.76±1.45	3.79±1.85	3.73±1.98
	AnchorOnly	–	5.01±1.26	4.52±0.94
	Proposed	–	**0.84±0.16**	0.45±0.11
	Proposed-CF	–	1.35±0.54	**0.39±0.12**
	Proposed-B	**3.51±1.47**	3.51±1.89	3.48±2.00
5	TargetOnly	–	4.79±0.96	4.56±0.86
	ProxyOnly	5.57±0.87	5.25±1.12	5.21±1.09
	IPW-Transport	3.33±1.45	3.43±1.52	3.59±1.74
	OM-Transport	3.24±1.46	3.25±1.58	3.35±1.75
	EntropyBal	3.53±1.47	3.48±1.52	3.56±1.74
	AnchorOnly	–	4.85±1.04	4.61±0.87
	Proposed	–	**0.78±0.16**	0.45±0.12
	Proposed-CF	–	1.16±0.34	**0.39±0.13**
	Proposed-B	**3.24±1.46**	2.81±1.43	3.26±1.77
10	TargetOnly	–	4.96±0.81	4.66±0.93
	ProxyOnly	5.64±0.90	5.53±0.94	5.41±1.37
	IPW-Transport	3.14±1.29	3.39±1.39	3.46±1.80
	OM-Transport	3.09±1.30	3.28±1.42	3.33±1.87
	EntropyBal	3.31±1.24	3.47±1.38	3.45±1.81
	AnchorOnly	–	5.04±0.85	4.71±0.94
	Proposed	–	**0.73±0.18**	0.45±0.14
	Proposed-CF	–	1.19±0.39	**0.39±0.15**
	Proposed-B	**3.09±1.30**	2.78±1.31	3.18±1.85
20	TargetOnly	–	4.78±0.77	4.47±0.83
	ProxyOnly	5.71±0.96	5.29±0.97	5.12±1.00
	IPW-Transport	2.86±1.19	2.88±1.38	3.02±1.56
	OM-Transport	**2.84±1.19**	2.83±1.39	2.91±1.58
	EntropyBal	2.99±1.16	2.96±1.36	3.03±1.55
	AnchorOnly	–	4.84±0.81	4.52±0.83
	Proposed	–	**0.69±0.16**	0.44±0.10
	Proposed-CF	–	1.08±0.31	**0.39±0.11**
	Proposed-B	**2.84±1.19**	**2.39±1.21**	2.63±1.53
50	TargetOnly	–	5.05±1.18	4.56±0.77
	ProxyOnly	5.55±0.85	5.62±1.50	5.25±1.02
	IPW-Transport	2.74±1.17	3.23±1.97	2.89±1.48
	OM-Transport	2.74±1.17	3.22±1.97	2.85±1.50
	EntropyBal	2.84±1.16	3.28±1.94	2.92±1.48
	AnchorOnly	–	5.10±1.24	4.62±0.78
	Proposed	–	**0.66±0.15**	0.45±0.16
	Proposed-CF	–	1.26±0.57	**0.41±0.16**
	Proposed-B	**2.74±1.17**	**2.24±1.61**	**2.38±1.51**

$p \in \{10, 20, 50, 100\}$ for `ProxyOnly`, transport baselines, and `Proposed-B`.

Key findings. (i) `Proposed-B` is best or near-best on PEHE across dimensions; (ii) transport baselines are competitive at higher p. Full results in Appendix C.

VI. Real-Life Data Experiments

To complement the synthetic experiments, we evaluate the proposed framework on a semi-synthetic benchmark built from the Infant Health and Development Program (IHDP) dataset [22]. IHDP provides real covariate distributions from an early-childhood intervention study while supplying known ground-truth treatment effects, enabling exact evaluation of CATE estimation accuracy.

A. The IHDP Dataset

The IHDP benchmark [22] comprises real baseline covariates from the Infant Health and Development Program: 25 features (19 binary, 6 continuous) per subject. Outcomes are semi-synthetically generated so that potential outcomes under control and treatment, $\mu_0(x)$ and $\mu_1(x)$, are known; the ground-truth CATE is $\tau(x) = \mu_1(x) - \mu_0(x)$. We use 50 standard realizations of the NPCI-style data, each with different outcome surfaces, to assess performance across heterogeneous effect structures while retaining repeatability.

TABLE IV
PEHE UNDER A5 VIOLATIONS: s/p=SPARSITY RATIO, Γ=NONLINEARITY. LOWER IS BETTER.

s/p	Method	Γ=0	0.5	1.0
0.05	TargetOnly	4.41±0.70	4.25±0.53	4.30±0.57
	ProxyOnly	4.97±0.89	4.72±0.68	4.84±0.72
	IPW-Transport	2.79±1.25	2.13±1.02	2.20±1.11
	OM-Transport	2.67±1.26	2.04±1.03	2.10±1.13
	EntropyBal	2.80±1.25	2.14±1.02	2.21±1.11
	AnchorOnly	4.45±0.70	4.28±0.53	4.34±0.56
	Proposed	0.44±0.11	**1.16±0.68**	**1.92±1.19**
	Proposed-CF	**0.38±0.12**	1.17±0.69	2.04±1.25
	Proposed-B	2.64±1.21	1.92±0.87	2.08±1.13
0.20	TargetOnly	4.80±0.75	4.32±0.55	4.43±0.60
	ProxyOnly	5.62±0.97	4.83±0.68	4.98±0.73
	IPW-Transport	4.18±1.29	2.53±0.89	2.69±1.02
	OM-Transport	4.07±1.31	2.46±0.91	2.56±1.06
	EntropyBal	4.19±1.29	2.54±0.90	2.71±1.01
	AnchorOnly	4.87±0.77	4.37±0.57	4.47±0.62
	Proposed	0.48±0.15	1.42±0.64	**2.41±1.18**
	Proposed-CF	**0.42±0.15**	**1.40±0.65**	2.57±1.22
	Proposed-B	4.03±1.34	2.43±0.95	2.55±1.06
1.00	TargetOnly	4.92±0.92	4.23±0.58	4.60±0.84
	ProxyOnly	5.75±1.21	4.77±0.71	5.04±0.94
	IPW-Transport	4.12±1.68	2.36±0.91	2.82±1.21
	OM-Transport	4.02±1.71	2.28±0.91	2.69±1.24
	EntropyBal	4.14±1.68	2.37±0.91	2.83±1.21
	AnchorOnly	4.99±0.94	4.29±0.60	4.63±0.84
	Proposed	0.45±0.13	1.29±0.64	**2.54±1.35**
	Proposed-CF	**0.39±0.13**	**1.28±0.64**	2.69±1.38
	Proposed-B	3.98±1.74	2.34±1.05	2.67±1.25

TABLE V
DISCONNECTED REGIME ($m_1 = 0$): ONLY TARGET PLACEBO AVAILABLE ($m_0 = 50$). METHODS REQUIRING TARGET TREATED DATA ARE UNDEFINED. MEAN$_{\pm\text{SD}}$ OVER 100 REPS. BEST IN BOLD.

(a) PEHE ↓

Method	p=10	p=20	p=50	p=100
ProxyOnly	2.34±0.67	3.37±0.84	5.78±1.22	8.21±1.23
IPW-Transport	1.80±0.83	2.24±1.10	3.26±1.83	4.10±2.05
OM-Transport	1.80±0.83	2.24±1.10	3.25±1.82	4.06±2.06
EntropyBal	1.81±0.83	2.25±1.10	3.40±1.79	4.77±1.93
Proposed-B	**1.63±0.75**	**1.95±0.92**	**3.25±1.82**	**4.05±2.06**

(b) ATE Error ↓

Method	p=10	p=20	p=50	p=100
ProxyOnly	**0.69±0.52**	0.85±0.66	1.33±1.04	1.86±1.60
IPW-Transport	0.77±0.56	0.80±0.60	0.76±0.59	1.00±0.89
OM-Transport	0.77±0.56	0.80±0.60	**0.74±0.55**	0.99±0.87
EntropyBal	0.77±0.56	0.79±0.59	0.76±0.58	**0.98±0.87**
Proposed-B	0.72±0.54	**0.73±0.56**	0.74±0.54	0.99±0.87

(c) ECE ↓

Method	p=10	p=20	p=50	p=100
ProxyOnly	0.93±0.48	1.07±0.56	1.56±0.92	2.19±1.43
IPW-Transport	0.84±0.55	0.92±0.58	0.95±0.62	1.19±0.88
OM-Transport	0.84±0.55	0.92±0.58	0.93±0.59	1.19±0.86
EntropyBal	0.85±0.55	0.92±0.58	0.99±0.63	1.43±0.93
Proposed-B	**0.79±0.53**	**0.84±0.56**	**0.93±0.59**	**1.19±0.86**

B. Semi-Synthetic Multi-Site Construction

We construct a multi-site setup from each IHDP realization by partitioning subjects into $C = 6$ clusters via k-means on the standardized covariates. One cluster is designated the target site; the remaining clusters form source sites. This induces natural covariate shift across sites while preserving known ground-truth potential outcomes. We subsample the target site to prescribed budgets (m_0, m_1) for the connected regime, or set $m_1 = 0$ and vary m_0 for the disconnected regime. In the disconnected setting we mask treated outcomes in the target during training and use target placebo outcomes only for screening-valid source selection, directly testing Assumption **A6**; evaluation remains fully supervised because τ is known on the target.

C. Evaluation Protocol

We use the same metrics as in §V: PEHE, absolute ATE error, Spearman rank correlation, policy regret, and ECE. Each scenario is run over 50 IHDP realizations (one run per realization). In the *connected* regime we sweep $(m_0, m_1) \in \{25, 50, 100\}^2$ and evaluate Option A methods: `TargetOnly`, `ProxyOnly`, `AnchorOnly`, transport baselines, and `Proposed` / `Proposed-CF`. In the *disconnected* regime we sweep $m_0 \in \{25, 50, 100, 200\}$ with $m_1 = 0$ and evaluate Option B methods: `ProxyOnly`, `IPW-Transport`, `OM-Transport`, `EntropyBal`, and `Proposed-B`.

D. Results and Interpretation

Connected regime ($m_1 > 0$). Table VI reports PEHE (mean $\pm$ SD over 50 realizations) across target budgets (m_0, m_1). `Proposed` is best or near-best in most cells, with clear gains over `ProxyOnly` and `TargetOnly` at small target sizes; `Proposed-CF` is competitive.

Disconnected regime ($m_1 = 0$). Table VII reports PEHE for the disconnected regime. Disconnected IHDP ($m_1 = 0$) is non-identified from target data. `Proposed-B` should therefore be interpreted as a working-model screen–then–transport estimator under Assumption A6, with irreducible cross-arm transport error. In this benchmark, `Proposed-B` is competitive on PEHE and strong on ranking; full metrics (ATE error, Spearman, regret, ECE) for both regimes are in Appendix C.

VII. Conclusion

Trial evidence is often abundant but hard to apply to the population of interest: sites differ, covariates shift, baseline risk drifts, and the evidence network can be weakly connected or disconnected. Yet clinical decisions still demand target-specific, patient-level counterfactual predictions.

We introduced a placebo-anchored proxy–gold framework that treats source-trial IPD outcomes as low-variance *proxy* signals and target placebo outcomes as scarce but high-fidelity *gold* calibration for baseline risk. This leads to a simple operating principle: anchor (and, in disconnected settings, screen) using placebo information, then use orthogonalized DR learning to stabilize effect estimation whenever randomization permits.

Our findings clarify that guarantees depend on the regime: connected targets (Option A) yield identified, orthogonal DR CATE estimates; disconnected targets (Option B) are non-identified and `Proposed-B` is a working-model screen–then–transport estimator under A6. Across synthetic and IHDP experiments, the proposed method is best or near-best in the connected regime and competitive in the disconnected regime (Section VI), with smooth degradation as assumptions weaken. The framework turns fragmented multi-trial evidence into calibrated, patient-level predictions while making transport assumptions explicit.

Several directions are immediate. First, disconnected-target reliability hinges on screening-valid transportability; diagnostics and sensitivity analyses around Assumption **A6** are important for deployment. Second, extending beyond GLMs to survival/time-to-event endpoints and richer nuisance learners would broaden applicability. Third, improving screening

TABLE VI
IHDP CONNECTED REGIME: PEHE ACROSS TARGET BUDGETS (MEAN $\pm$ SD, 50 REALIZATIONS).

$m_0 = 25$ Method	m_1=25	m_1=50	m_1=100
TargetOnly	4.22±6.07	5.45±7.12	3.09±3.97
ProxyOnly	3.33±4.71	4.21±5.59	2.36±2.75
AnchorOnly	3.12±4.32	3.88±4.87	2.37±2.57
IPW-Transport	3.69±6.23	3.76±5.55	2.29±2.21
EntropyBal	4.87±8.67	6.41±13.96	3.55±4.33
OM-Transport	3.40±5.73	3.46±4.71	2.05±1.98
Proposed	**2.46±4.36**	**2.88±3.78**	**1.57±1.92**
Proposed-CF	3.55±6.51	3.89±4.82	2.47±2.74
$m_0 = 50$ Method	**m_1=25**	**m_1=50**	**m_1=100**
TargetOnly	3.39±5.17	3.63±5.43	4.58±6.19
ProxyOnly	3.34±5.30	3.22±4.70	3.90±5.21
AnchorOnly	2.85±4.29	2.96±4.20	3.54±4.78
IPW-Transport	3.72±6.55	3.30±4.75	4.84±7.40
EntropyBal	4.31±6.70	4.07±6.15	9.30±18.58
OM-Transport	3.51±6.35	3.10±4.75	3.99±5.71
Proposed	**2.51**±4.64	**2.42±4.62**	**2.98±4.75**
Proposed-CF	3.12±4.76	2.92±4.64	3.38±4.87
$m_0 = 100$ Method	**m_1=25**	**m_1=50**	**m_1=100**
TargetOnly	2.48±3.33	3.11±4.86	2.97±4.10
ProxyOnly	3.04±4.11	3.38±5.22	3.21±4.38
AnchorOnly	2.64±3.85	2.89±4.51	2.77±3.93
IPW-Transport	3.25±4.53	3.76±5.95	3.82±5.09
EntropyBal	4.04±5.45	5.35±8.40	4.64±6.75
OM-Transport	3.22±4.86	3.63±6.12	3.59±5.13
Proposed	**2.06**±3.31	**2.41**±4.24	**2.31±3.60**
Proposed-CF	2.32±2.99	2.62±4.34	2.57±3.75

TABLE VII
IHDP DISCONNECTED REGIME ($m_1 = 0$): PEHE ACROSS PLACEBO BUDGETS (MEAN $\pm$ SD, 50 REALIZATIONS).

Method	25	50	100	200
ProxyOnly	2.64±2.65	3.46±4.88	3.50±4.27	3.79±4.53
IPW-Transport	2.30±3.08	3.36±5.84	4.87±6.95	4.79±6.95
EntropyBal	2.82±3.02	3.61±5.65	5.47±7.99	5.63±8.62
OM-Transport	2.28±3.41	3.37±6.43	4.18±5.96	4.50±6.64
Proposed-B	**2.11**±3.44	**3.19**±6.23	**3.90**±5.88	**4.33**±6.59

(e.g., multi-endpoint screening or representation learning) may reduce reliance on placebo alone. Finally, uncertainty quantification in small-gold regimes remains central, and combining influence-function variance estimates with resampling or calibration is promising.

Overall, the placebo-anchored viewpoint separates what is *identified* in connected targets from what is *transported* in disconnected targets, while delivering calibrated outputs suitable for individualized decision analysis.

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APPENDIX

A. Asymptotics and Error Decompositions

There are two regimes:

1) **Connected target (Option A / `Proposed-CF`).** The target site has both arms. We estimate $\mu_{0,0}$ and $\mu_{1,0}$ via `glmtrans` (with automatic source detection) and then run a cross-fitted DR learner on the *target* to obtain $\widehat{\tau}_{\mathrm{DR}}(x)$.
2) **Disconnected target (Option B / `Proposed-B`).** The target has placebo only. We use the target placebo arm solely to *screen* transferable sources, then learn a DR CATE model on the selected sources and transport it to the target by prediction. In this regime, $\tau_0(x)$ is not nonparametrically identified from target data; we therefore provide a *transport* error decomposition under Assumption **A6**.

Throughout we adopt the high-dimensional GLM transfer-learning theory of [15], which underpins (i) the error rates of the arm-specific `glmtrans` outcome regressions and (ii) the consistency of the automatic transferable-source detection.

B. Setup and Notation

Let $S \in \{0, 1, \ldots, C\}$ index sites, with $S = 0$ the target and $S = c \geq 1$ sources. For each unit observe $O = (X, A, Y, S)$ with $A \in \{0, 1\}$ and potential outcomes $(Y(0), Y(1))$. Define site- and arm-specific outcome regressions and CATEs

$$\mu_{a,c}(x) := \mathbb{E}[Y \mid A = a, X = x, S = c],$$
$$\tau_c(x) := \mu_{1,c}(x) - \mu_{0,c}(x).$$

The target CATE is $\tau_0(x)$.

Within each site c, the trial is randomized with known propensity $e_c(x) = \mathbb{P}(A = 1 \mid X = x, S = c)$. Assumption **A3** implies $\epsilon \leq e_c(x) \leq 1 - \epsilon$ on the relevant support.

C. Stage 1 Nuisance Rates via [15] (`glmtrans`)

Fix an arm $a \in \{0, 1\}$. The `glmtrans` procedure outputs an estimator $\widehat{\beta}_{a,0}$ for the target-arm GLM parameter (and thus $\widehat{\mu}_{a,0}(x)$), using: (i) automatic transferable-source detection (Trans-GLM; Algorithm 2 of [15]), followed by (ii) two-step transfer estimation on the selected sources (Ah-Trans-GLM; Algorithm 1).

To connect to the DR analysis, we summarize the implication we need: an $L_2(P_0)$ prediction rate for $\widehat{\mu}_{a,0}$ on the target covariate distribution P_0.

Lemma 1 (Prediction error rate for `glmtrans` outcome regressions)**.** *Assume the high-dimensional GLM conditions of [15] for the arm-a regression (including bounded design/curvature, sparsity, and the transfer-heterogeneity condition indexed by h). Let $\widehat{\mu}_{a,0}$ be the `glmtrans` estimator using automatic source detection. Then with probability at least $1 - \delta$,*

$$\|\widehat{\mu}_{a,0} - \mu_{a,0}\|_{L_2(P_0)} \lesssim \underbrace{\left(\frac{s \log p}{n_{a,0} + n_{a,\mathcal{A}_a}}\right)^{1/2}}_{\text{transfer (variance reduction)}} + \underbrace{\left(\frac{\log p}{n_{a,0}}\right)^{1/4} h^{1/2}}_{\text{target debiasing term}},$$

where $n_{a,0}$ is the target arm-a sample size and $n_{a,\mathcal{A}_a}$ is the total sample size of the selected transferable sources for arm a. Moreover, the automatic detection step satisfies $\widehat{\mathcal{A}}_a = \mathcal{A}_{a,h}$ with probability at least $1 - \delta$ under the detection-consistency conditions of [15] (Theorem 4).

Remark. Lemma 1 is a direct specialization of the '1/'2 estimation-error and detection results in [15] (Theorems 1 and 4), translated into prediction error on P_0. The key point for our appendix is that, when sufficiently many transferable source samples are available, $\widehat{\mu}_{a,0}$ can achieve a strictly faster rate than target-only Lasso under the same sparsity.

D. Option A (Connected Target): DR Orthogonalization on the Target

a) Estimator.: On the target site $S = 0$, using cross-fitted nuisance regressions $\widehat{\mu}_{0,0}, \widehat{\mu}_{1,0}$, form the DR pseudo-outcome for each target unit i:

$$\psi_i := \widehat{\mu}_{1,0}(X_i) - \widehat{\mu}_{0,0}(X_i) \tag{2}$$
$$+ \frac{A_i - e_0(X_i)}{e_0(X_i)\{1 - e_0(X_i)\}}\Big(Y_i - \widehat{\mu}_{A_i,0}(X_i)\Big), \qquad (S_i = 0). \tag{3}$$

We then fit a second-stage regression $\widehat{\tau}_{\mathrm{DR}}(\cdot)$ of ψ on X on the target. For the asymptotic statement below, we treat the second stage as a linear/sieve regression with a fixed (or slowly-growing) feature map $b(X)$, so that pointwise inference at a fixed x is well-defined.

b) Working target.: In the connected target, $\tau_0(x)$ is identified by randomization and is the natural target.

Theorem 1 (Asymptotic linearity for Option A (`Proposed-CF`))**.** *Fix a covariate value x and suppose the second-stage regression operator is linear in a feature map $b(\cdot)$ with stable Gram matrix. Assume (i) Assumptions **A1–A3**, (ii) finite fourth moments of Y and bounded propensities, (iii) cross-fitting of nuisances, and (iv) nuisance rates*

$$\begin{aligned}\|\widehat{\mu}_{0,0}-\mu_{0,0}\|_{L_2(P_0)} &= o_p(n_0^{-1/4}),\\ \|\widehat{\mu}_{1,0}-\mu_{1,0}\|_{L_2(P_0)} &= o_p(n_0^{-1/4}),\end{aligned} \tag{4}$$

where n_0 is the target sample size (both arms). Then the Option A CATE regression admits the pointwise expansion

$$\widehat{\tau}_{\mathrm{DR}}(x)-\tau_0(x)=\frac{1}{n_0}\sum_{i:S_i=0}\phi(O_i;x)+o_p(n_0^{-1/2}),$$

for an influence function $\phi(\cdot;x)$ with $\mathbb{E}[\phi(O;x)\mid S=0]=0$ and $\mathbb{E}[\phi(O;x)^2\mid S=0]<\infty$. Consequently,

$$\sqrt{n_0}\big(\widehat{\tau}_{\mathrm{DR}}(x)-\tau_0(x)\big)\ \Rightarrow\ \mathcal{N}(0,\,V(x)),$$

for $V(x):=\mathrm{Var}(\phi(O;x)\mid S=0)$.

By Lemma 1, the nuisance rate condition (4) holds whenever the `glmtrans` prediction error on P_0 is $o_p(n_0^{-1/4})$ for each arm. This is achievable under the regimes in [15] where transferable sources exist and $n_{a,\mathcal{A}_a}$ is sufficiently large relative to $(s,p,n_{a,0})$.

c) Second-order remainder (explicit product structure).: Under the orthogonality of the DR score and cross-fitting, nuisance errors enter only through a second-order term:

$$\begin{aligned}\widehat{\tau}_{\mathrm{DR}}(x)-\tau_0(x)=&\frac{1}{n_0}\sum_{i:S_i=0}\phi(O_i;x)\\ &+O_p\Big(\|\widehat{\mu}_{0,0}-\mu_{0,0}\|_{L_2(P_0)}\cdot\|\widehat{\mu}_{1,0}-\mu_{1,0}\|_{L_2(P_0)}\Big)\\ &+o_p(n_0^{-1/2}).\end{aligned}$$

Since propensities are known by design, there is no additional treatment-model error term.

E. Option B (Disconnected Target): Transported Source-DR and Structural Bias under A6

In the disconnected regime, the target has placebo only, so $\mu_{1,0}$ and τ_0 are not identified from target data alone. Our implemented `Proposed-B` procedure is:

1) **Placebo-based screening.** Apply Trans-GLM detection on $\mathcal{D}_{0,0}=\{(X,Y):S=0,A=0\}$ to obtain $\widehat{\mathcal{C}}_0$.
2) **Source DR learning.** Pool selected sources $\mathcal{D}^\star=\bigcup_{c\in\widehat{\mathcal{C}}_0}\mathcal{D}_c$ (both arms observed), fit source outcome regressions, compute source DR pseudo-outcomes, and learn a CATE function $\widehat{\tau}_B(\cdot)$ on $\mathcal{D}^\star$.
3) **Transport.** Output $\widehat{\tau}_B(x)$ for target units by prediction.

Assumption **A6** formalizes that placebo-based screening yields a set of sources whose CATEs are close to the target CATE on the target covariate support. Accordingly, the appropriate guarantee is a *transport* error decomposition (estimation + structural bias), rather than a DR CLT on target data.

Theorem 2 (Transport error decomposition for Option B (`Proposed-B`))**.** *Assume Assumptions **A1–A3** and Assumption **A6**. Let $\mathcal{C}_0^\star$ be the subset in Assumption **A6** and let $\widehat{\mathcal{C}}_0$ be the detected subset from target placebo. Define the* oracle transported target *as the source-mixture CATE*

$$\tau^\star(x):=\mathbb{E}[\tau_S(x)\mid S\in\mathcal{C}_0^\star]\,,$$

and let $\widehat{\tau}_B$ be the learned source-DR predictor trained on $\widehat{\mathcal{C}}_0$ and evaluated on target covariates. Then, on the target covariate distribution P_0,

$$\|\widehat{\tau}_B-\tau_0\|_{L_2(P_0)}\ \le\ \underbrace{\|\widehat{\tau}_B-\tau^\star\|_{L_2(P_0)}}_{\textit{estimation on selected sources}} \tag{5}$$

$$+\ \underbrace{\|\tau^\star-\tau_0\|_{L_2(P_0)}}_{\textit{structural transport bias}}+\underbrace{\Delta_{\mathrm{sel}}}_{\textit{screening error}}\quad. \tag{6}$$

*Moreover, Assumption **A6**(b) implies $\|\tau^\star-\tau_0\|_{L_2(P_0)}\le\epsilon_\tau$. If additionally $\mathbb{P}(\widehat{\mathcal{C}}_0\subseteq\mathcal{C}_0^\star)\ge 1-\eta$, then $\Delta_{\mathrm{sel}}\lesssim\eta^{1/2}$ (up to constants depending on bounded second moments of $\tau_S(X)$).*

Remark (what is provable in Option B). Theorem 2 makes explicit that Option B performance is governed by: (i) learning error on the selected sources (controlled by source sample sizes and the complexity of the CATE learner), (ii) the irreducible structural term ϵ_τ in Assumption **A6** (cross-arm non-transfer not detectable from placebo alone), and (iii) the screening error probability η (controlled by the detection consistency theory of [15] on the placebo arm).

F. Proofs

Proof of Theorem 1. Let $Z := b(X)$ and

$$G_0 := \mathbb{E}[ZZ^\top \mid S = 0].$$

By the stable-Gram assumption, $\lambda_{\min}(G_0) > 0$, and the sample Gram matrix

$$\widehat{G} := \frac{1}{n_0} \sum_{i:S_i=0} Z_i Z_i^\top$$

satisfies $\widehat{G}^{-1} = G_0^{-1} + o_p(1)$. For notational simplicity we write the proof in the fixed-dimensional correctly specified linear case, so that

$$\tau_0(x') = b(x')^\top \theta_0$$

for some coefficient vector θ_0 and all target covariate values x'. The slowly growing sieve case is handled identically after absorbing the approximation error into the final $o_p(n_0^{-1/2})$ remainder.

Define the oracle pseudo-outcome

$$\psi_i^0 := \mu_{1,0}(X_i) - \mu_{0,0}(X_i) + \frac{A_i - e_0(X_i)}{e_0(X_i)\{1 - e_0(X_i)\}}\Big(Y_i - \mu_{A_i,0}(X_i)\Big), \qquad (S_i = 0).$$

By Assumptions **A1–A3**,

$$\mathbb{E}[Y_i - \mu_{A_i,0}(X_i) \mid X_i, A_i, S_i = 0] = 0,$$

and therefore

$$\begin{aligned}\mathbb{E}[\psi_i^0 \mid X_i, S_i = 0] &= \mu_{1,0}(X_i) - \mu_{0,0}(X_i) + \frac{\mathbb{E}[A_i - e_0(X_i) \mid X_i, S_i = 0]}{e_0(X_i)\{1 - e_0(X_i)\}} \cdot 0 \\ &= \tau_0(X_i).\end{aligned} \tag{7}$$

Thus the population normal equations for the second-stage regression are

$$\mathbb{E}\big[Z\big(\psi^0 - Z^\top\theta_0\big) \mid S = 0\big] = 0.$$

Let $\widehat{\psi}_i$ denote the cross-fitted pseudo-outcome in (3), and let

$$\widehat{\theta} := \widehat{G}^{-1}\Big(\frac{1}{n_0}\sum_{i:S_i=0} Z_i \widehat{\psi}_i\Big)$$

be the second-stage coefficient vector, so that $\widehat{\tau}_{\mathrm{DR}}(x) = b(x)^\top\widehat{\theta}$. Then

$$\begin{aligned}\widehat{\theta} - \theta_0 &= \widehat{G}^{-1}\Big(\frac{1}{n_0}\sum_{i:S_i=0} Z_i(\widehat{\psi}_i - Z_i^\top\theta_0)\Big) \\ &= \underbrace{\widehat{G}^{-1}\Big(\frac{1}{n_0}\sum_{i:S_i=0} Z_i(\psi_i^0 - Z_i^\top\theta_0)\Big)}_{=:T_{1,n}} + \underbrace{\widehat{G}^{-1}\Big(\frac{1}{n_0}\sum_{i:S_i=0} Z_i(\widehat{\psi}_i - \psi_i^0)\Big)}_{=:T_{2,n}}.\end{aligned} \tag{8}$$

Step 1: the generated-pseudo-outcome term is negligible. Fix a cross-fitting fold k and an observation i in the corresponding evaluation fold. Write

$$\Delta_a^{(-k)}(x) := \widehat{\mu}_{a,0}^{(-k)}(x) - \mu_{a,0}(x), \qquad a \in \{0, 1\}.$$

Then

$$\widehat{\psi}_i - \psi_i^0 = \Delta_1^{(-k)}(X_i) - \Delta_0^{(-k)}(X_i) - \frac{A_i - e_0(X_i)}{e_0(X_i)\{1 - e_0(X_i)\}}\Delta_{A_i}^{(-k)}(X_i).$$

Because the nuisance fits for fold k are trained only on the complementary sample, the functions $\Delta_a^{(-k)}$ are independent of the evaluation observations in that fold. Conditioning on the training sample and on X_i, and using

$$\mathbb{E}\left[\frac{A_i}{e_0(X_i)} \;\middle|\; X_i, S_i = 0\right] = 1,$$
$$\mathbb{E}\left[\frac{1-A_i}{1-e_0(X_i)} \;\middle|\; X_i, S_i = 0\right] = 1,$$

we obtain

$$\mathbb{E}[\widehat{\psi}_i - \psi_i^0 \mid X_i, \mathcal{I}_{-k}, S_i = 0] = \Delta_1^{(-k)}(X_i) - \Delta_0^{(-k)}(X_i) - \Delta_1^{(-k)}(X_i) + \Delta_0^{(-k)}(X_i) = 0. \tag{9}$$

Moreover, bounded propensities imply

$$\mathbb{E}\Big[(\widehat{\psi}_i - \psi_i^0)^2 \mid \mathcal{I}_{-k}, S_i = 0\Big] \lesssim \|\Delta_0^{(-k)}\|^2_{L_2(P_0)} + \|\Delta_1^{(-k)}\|^2_{L_2(P_0)}.$$

Since Z has finite second moments, the conditional variance of the empirical average in $T_{2,n}$ is therefore bounded by

$$\mathbb{E}\left[\Big\|\frac{1}{n_0}\sum_{i:S_i=0} Z_i(\widehat{\psi}_i - \psi_i^0)\Big\|^2 \;\middle|\; \{\mathcal{I}_{-k}\}\right] \lesssim \frac{\|\widehat{\mu}_{0,0} - \mu_{0,0}\|^2_{L_2(P_0)} + \|\widehat{\mu}_{1,0} - \mu_{1,0}\|^2_{L_2(P_0)}}{n_0}.$$

Hence

$$\begin{aligned}\Big\|\frac{1}{n_0}\sum_{i:S_i=0} Z_i(\widehat{\psi}_i - \psi_i^0)\Big\| &= O_p\bigg(\frac{\|\widehat{\mu}_{0,0} - \mu_{0,0}\|_{L_2(P_0)} + \|\widehat{\mu}_{1,0} - \mu_{1,0}\|_{L_2(P_0)}}{\sqrt{n_0}}\bigg)\\ &= o_p(n_0^{-1/2}).\end{aligned}$$

because (4) implies each nuisance error is $o_p(1)$. Since $\widehat{G}^{-1} = O_p(1)$, we conclude that

$$T_{2,n} = o_p(n_0^{-1/2}).$$

Step 2: asymptotic linearity of the oracle term. By (7),

$$\psi_i^0 - Z_i^\top \theta_0 = \frac{A_i - e_0(X_i)}{e_0(X_i)\{1 - e_0(X_i)\}}\Big(Y_i - \mu_{A_i,0}(X_i)\Big).$$

Therefore

$$T_{1,n} = \widehat{G}^{-1}\Big(\frac{1}{n_0}\sum_{i:S_i=0} Z_i \frac{A_i - e_0(X_i)}{e_0(X_i)\{1 - e_0(X_i)\}}\big(Y_i - \mu_{A_i,0}(X_i)\big)\Big).$$

Evaluating at the fixed covariate value x and using $\widehat{G}^{-1} = G_0^{-1} + o_p(1)$ gives

$$\begin{aligned} b(x)^\top T_{1,n} &= \frac{1}{n_0}\sum_{i:S_i=0} b(x)^\top G_0^{-1} Z_i \frac{A_i - e_0(X_i)}{e_0(X_i)\{1 - e_0(X_i)\}}\big(Y_i - \mu_{A_i,0}(X_i)\big) + o_p(n_0^{-1/2})\\ &= \frac{1}{n_0}\sum_{i:S_i=0} \phi(O_i; x) + o_p(n_0^{-1/2}), \end{aligned}\tag{10}$$

where

$$\phi(O_i; x) := b(x)^\top G_0^{-1} Z_i \frac{A_i - e_0(X_i)}{e_0(X_i)\{1 - e_0(X_i)\}}\big(Y_i - \mu_{A_i,0}(X_i)\big).$$

Equation (7) implies

$$\mathbb{E}[\phi(O; x) \mid S = 0] = 0.$$

Finite fourth moments of Y, bounded propensities, and finite second moments of Z imply

$$\mathbb{E}[\phi(O; x)^2 \mid S = 0] < \infty.$$

Hence the classical central limit theorem yields

$$\sqrt{n_0}\,\frac{1}{n_0}\sum_{i:S_i=0} \phi(O_i; x) \Rightarrow \mathcal{N}(0, V(x)), \qquad V(x) := \mathrm{Var}(\phi(O; x) \mid S = 0).$$

Step 3: conclusion. Combining (8), the negligibility of $T_{2,n}$, and (10), we obtain

$$\widehat{\tau}_{\mathrm{DR}}(x) - \tau_0(x) = \frac{1}{n_0} \sum_{i:S_i=0} \phi(O_i; x) + o_p(n_0^{-1/2}),$$

which is the desired asymptotic linear representation. The stated normal limit follows immediately.

Finally, the orthogonality calculation in (9) shows that the first-order Gateaux derivative of the target moment with respect to each nuisance regression vanishes at the truth. Thus the deterministic plug-in bias is at least quadratic in the nuisance errors, which is the source of the product-form remainder displayed before this proof. □

Proof of Theorem 2. Let

$$E := \{\widehat{\mathcal{C}}_0 \subseteq \mathcal{C}_0^\star\}$$

denote the screening-success event. Introduce the auxiliary predictor

$$\widetilde{\tau}_B := \widehat{\tau}_B \, \mathbf{1}_E + \tau^\star \, \mathbf{1}_{E^c}.$$

By the triangle inequality,

$$\|\widehat{\tau}_B - \tau_0\|_{L_2(P_0)} \leq \|\widehat{\tau}_B - \widetilde{\tau}_B\|_{L_2(P_0)} + \|\widetilde{\tau}_B - \tau^\star\|_{L_2(P_0)} + \|\tau^\star - \tau_0\|_{L_2(P_0)}. \tag{11}$$

Now

$$\widetilde{\tau}_B - \tau^\star = (\widehat{\tau}_B - \tau^\star)\mathbf{1}_E,$$

so

$$\|\widetilde{\tau}_B - \tau^\star\|_{L_2(P_0)} \leq \|\widehat{\tau}_B - \tau^\star\|_{L_2(P_0)}.$$

Define

$$\Delta_{\mathrm{sel}} := \|\widehat{\tau}_B - \widetilde{\tau}_B\|_{L_2(P_0)} = \|(\widehat{\tau}_B - \tau^\star)\mathbf{1}_{E^c}\|_{L_2(P_0)}.$$

Substituting these identities into (11) yields

$$\|\widehat{\tau}_B - \tau_0\|_{L_2(P_0)} \leq \|\widehat{\tau}_B - \tau^\star\|_{L_2(P_0)} + \|\tau^\star - \tau_0\|_{L_2(P_0)} + \Delta_{\mathrm{sel}},$$

which is (6).

To control the structural transport term, note that

$$\tau^\star(x) - \tau_0(x) = \mathbb{E}[\tau_S(x) - \tau_0(x) \mid S \in \mathcal{C}_0^\star].$$

Applying Minkowski's inequality and then Assumption **A6**(b),

$$\begin{aligned}
\|\tau^\star - \tau_0\|_{L_2(P_0)} &\leq \mathbb{E}\big[\|\tau_S - \tau_0\|_{L_2(P_0)} \mid S \in \mathcal{C}_0^\star\big] \\
&\leq \epsilon_\tau.
\end{aligned}$$

Finally, suppose $\mathbb{P}(E) \geq 1 - \eta$, so that $\mathbb{P}(E^c) \leq \eta$. If

$$M_2 := \mathbb{E}\Big[\|\widehat{\tau}_B - \tau^\star\|_{L_2(P_0)}^2\Big] < \infty,$$

then Cauchy–Schwarz gives

$$\begin{aligned}
\mathbb{E}[\Delta_{\mathrm{sel}}] &= \mathbb{E}\big[\|\widehat{\tau}_B - \tau^\star\|_{L_2(P_0)} \mathbf{1}_{E^c}\big] \\
&\leq \Big(\mathbb{E}\Big[\|\widehat{\tau}_B - \tau^\star\|_{L_2(P_0)}^2\Big]\Big)^{1/2} \mathbb{P}(E^c)^{1/2} \\
&\leq M_2^{1/2} \eta^{1/2}.
\end{aligned}$$

Therefore $\Delta_{\mathrm{sel}} = O_p(\eta^{1/2})$ by Markov's inequality. The constant $M_2^{1/2}$ is finite whenever the transported source CATEs (and hence the source-DR learner trained on them) have bounded second moments on the target support, which is exactly the moment condition invoked in the theorem. □

A. Data-Generating Process Details

The synthetic data generator (`FairSyntheticRCTGenerator`) creates multi-site RCT data with the following structure.

a) Covariate Generation.: For each site $c \in \{0, 1, \ldots, C\}$ (where $c = 0$ is the target), covariates are generated as:

$$X_i \mid c \sim \mathcal{N}(\mu_c, I_p),$$

where μ_c controls site-level covariate shift. The target site mean is:

$$\mu_0 = (1 - \lambda_{\text{overlap}}) \cdot \bar{\mu}_{\text{source}} + \lambda_{\text{overlap}} \cdot \Delta,$$

where $\lambda_{\text{overlap}} \in [0, 1]$ controls overlap (0 = perfect overlap, 1 = maximal shift), and Δ is a random shift vector. Default: $\lambda_{\text{overlap}} = 0.25$ giving source-vs-target classification AUC ≈ 0.75.

b) Treatment Assignment.: Treatment is randomized via logistic propensity:

$$A_i \mid X_i \sim \text{Bernoulli}(e(X_i)), \quad e(X_i) = \text{expit}(X_i^\top \gamma_e),$$

where γ_e is sparse with 3 active coefficients. For the target site, we set $e(X) \equiv p_T$ to achieve the desired (m_0, m_1) split.

c) Outcome Model.: Outcomes follow the proxy–deviation decomposition:

$$Y_i = \mu_{A_i}^{\text{proxy}}(X_i) + \delta_{A_i, c_i}(X_i) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2),$$

where:

- $\mu_a^{\text{proxy}}(x) = x^\top \beta_a + \alpha_a$ is the shared (transferable) component
- $\delta_{a,c}(x) = x^\top \gamma_{a,c}$ is the sparse site-specific deviation
- $\gamma_{a,c}$ has $\lfloor s \cdot p \rfloor$ nonzero entries (sparsity ratio s/p)

d) Nontransfer Scale.: The nontransfer scale σ_ν controls $\|\gamma_{a,c}\|_2$ relative to $\|\beta_a\|_2$. Default: $\sigma_\nu = 0.1$, giving signal-to-noise ratio (SNR) $\approx$ 3–4 for the transferable component.

e) Nonlinearity Violations (A5 Sensitivity).: For testing A5 violations, we replace the linear deviation with a mixture:

$$\delta_{a,c}(x) = (1 - \lambda) \cdot x^\top \gamma_{a,c} + \lambda \cdot g(x),$$

where $g(x) = \sum_{j=1}^{p} \tanh(x_j)$ is a smooth additive nonlinearity and $\lambda \in \{0, 0.5, 1\}$ controls the mixture.

f) Sample Sizes.:

- Source sites: $C \in \{2, 5, 10, 20, 50\}$ sites, 1000 samples per site (2000 per site in some sweeps)
- Target site: (m_0, m_1) ranges from $(50, 0)$ to $(550, 500)$
- Dimensions: $p \in \{10, 20, 50, 100\}$

B. Benchmark Method Implementations

All methods receive the same data and are evaluated on held-out target samples. We describe each method category below.

1) Baseline Methods:

a) TargetOnly (TargetOnlyDR).: Standard doubly robust (DR) learner using only target data. Fits outcome models $\hat{\mu}_0, \hat{\mu}_1$ on target placebo and treated samples, then computes DR pseudo-outcomes:

$$\tilde{\tau}_i = \frac{A_i(Y_i - \hat{\mu}_1(X_i))}{\hat{e}(X_i)} - \frac{(1 - A_i)(Y_i - \hat{\mu}_0(X_i))}{1 - \hat{e}(X_i)} + \hat{\mu}_1(X_i) - \hat{\mu}_0(X_i).$$

Requires target treated data ($m_1 > 0$). Uses Ridge regression for $\hat{\mu}_a$.

b) ProxyOnly.: Pools all source data to fit outcome models $\hat{\mu}_a^{\text{proxy}}(x)$ using Random Forest, then predicts CATE as $\hat{\tau}(x) = \hat{\mu}_1^{\text{proxy}}(x) - \hat{\mu}_0^{\text{proxy}}(x)$. Does not use any target data for calibration—serves as a transfer-without-anchoring baseline.

2) Transport Baselines:

a) IPW-Transport (IPWTransport).: Inverse probability weighting estimator for transportability [23]. Fits a selection model $\hat{s}(X) = P(\text{source} \mid X)$ via logistic regression, then reweights source outcomes:

$$\hat{\tau}^{\text{IPW}} = \frac{1}{n_1} \sum_{i: A_i = 1} \frac{Y_i \cdot \hat{w}(X_i)}{\sum_j \hat{w}(X_j)} - \frac{1}{n_0} \sum_{i: A_i = 0} \frac{Y_i \cdot \hat{w}(X_i)}{\sum_j \hat{w}(X_j)},$$

where $\hat{w}(x) = (1 - \hat{s}(x))/\hat{s}(x)$ are the odds weights.

b) OM-Transport (OutcomeModelTransport).: Outcome model transport estimator [8]. Fits arm-specific outcome models $\hat{g}_a(x)$ on source data, then averages predictions over target covariates:

$$\hat{\tau}^{\text{OM}} = \frac{1}{m_0 + m_1} \sum_{i \in \text{target}} \left[\hat{g}_1(X_i) - \hat{g}_0(X_i)\right].$$

Uses Random Forest for $\hat{g}_a$.

c) EntropyBal (EntropyBalancing).: Entropy balancing for covariate shift adjustment [13]. Finds weights w_i that minimize $\sum_i w_i \log(w_i)$ subject to moment constraints:

$$\sum_{i \in \text{source}} w_i X_i = \bar{X}_{\text{target}}.$$

Then applies weighted outcome regression.

3) Anchor Methods:

a) AnchorOnly.: Placebo-anchored outcome models with DR Stage 3. Fits proxy models on source, then learns sparse corrections $\hat{\delta}_a(x) = x^\top \hat{\gamma}_a$ using target placebo residuals. Applies DR correction using target treated data.

4) Proposed Methods (glmtrans-based): Our proposed method uses the `glmtrans` R package [24] for ℓ_1-penalized transfer learning with automatic source detection.

a) Proposed (Glmtrans_Auto).: Two-step transfer learning:

1) **Source detection**: For each source k, test $H_0 : \beta^{(k)} = \beta^{(0)}$ using truncated ℓ_1 estimator. Select transferable sources $\mathcal{A}$.
2) **Transfer estimation**: Solve

$$\hat{\beta}_a = \arg\min_{\beta} \frac{1}{2m} \|Y_a - X_a \beta\|_2^2 + \lambda_1 \|\beta - \bar{\beta}_{\mathcal{A}}\|_1 + \lambda_2 \|\beta\|_1,$$

where $\bar{\beta}_{\mathcal{A}}$ is the pooled source estimate from selected sources.

CATE is computed as plug-in: $\hat{\tau}(x) = x^\top (\hat{\beta}_1 - \hat{\beta}_0)$.

b) Proposed-CF (Glmtrans_DR_CrossFit).: Combines glmtrans with cross-fitted DR pseudo-outcomes:

1) Split target data into $K = 5$ folds
2) For each fold k: fit $\hat{\mu}_0^{(-k)}, \hat{\mu}_1^{(-k)}, \hat{e}^{(-k)}$ on remaining folds
3) Compute DR pseudo-outcomes $\tilde{\tau}_i$ for fold k samples
4) Run glmtrans on pseudo-outcomes

c) Proposed-B (Glmtrans_OptionB).: For disconnected targets ($m_1 = 0$):

1) Run glmtrans source detection on source data only
2) Learn transfer operator $\hat{M}$ relating placebo and treated coefficients across sources
3) Apply $\hat{M}$ to transfer placebo-anchored predictions to treated arm
4) Compute CATE using source-based DR (no target treated needed)

C. Hyperparameters and Tuning

a) Regularization.:

- LASSO/Ridge: 5-fold cross-validation with `LassoCV`/`RidgeCV`
- glmtrans: λ selected by K-fold cross-validation minimizing MSE (`lambda.min` in `glmnet`); $K = 5$ for CATE and Option B, $K = 10$ for the main transfer fit. Covariates are standardized by `glmnet` before fitting; coefficients can be back-transformed to the original scale for interpretability.
- Random Forest: 100 trees, max depth 10, min samples leaf 5

b) Cross-fitting.: DR methods use $K = 5$ stratified folds (stratified by treatment arm within each fold).

c) Propensity Clipping.: Propensity scores are clipped to $[\epsilon, 1 - \epsilon]$ with $\epsilon = 0.01$ to avoid extreme weights.

d) Reproducibility.: All experiments use fixed random seeds. Each scenario runs $R = 100$ Monte Carlo replicates. Code is available at https://github.com/wangzilongri/SparseTLDRICHI2026.

The following tables report full results with error bars (mean ± SD over 100 MC replicates) for all metrics across all experiments. Method abbreviations: Target=TargetOnly, Proxy=ProxyOnly, IPW=IPW-Transport, OM=OM-Transport, EB=EntropyBal, Anch=AnchorOnly, Prop=Proposed, Prop-CF=Proposed-CF, Prop-B=Proposed-B.

A. Target Budget vs. Dimensionality

TABLE VIII
PEHE (MEAN±SD) ACROSS BUDGET AND p. LOWER IS BETTER.

p	Method	(50,0)	(100,50)	(150,100)	(250,200)	(550,500)
10	TargetOnly	–	1.69±0.49	1.43±0.39	1.42±0.38	1.38±0.45
	ProxyOnly	2.34±0.67	2.09±0.66	1.96±0.66	2.08±0.68	2.10±0.83
	IPW-Transport	1.80±0.83	1.80±0.75	1.60±0.70	1.78±0.77	1.82±1.06
	OM-Transport	1.80±0.83	1.81±0.75	1.60±0.70	1.78±0.77	1.81±1.06
	EntropyBal	1.81±0.83	1.81±0.75	1.60±0.70	1.78±0.77	1.82±1.06
	AnchorOnly	–	1.71±0.53	1.46±0.41	1.44±0.41	1.39±0.46
	Proposed	–	**0.53±0.14**	0.43±0.16	0.35±0.15	0.28±0.14
	Proposed-CF	–	0.54±0.15	**0.41±0.15**	**0.34±0.16**	**0.28±0.14**
	Proposed-B	**1.63±0.75**	1.73±0.74	1.51±0.66	1.72±0.74	1.81±1.06
20	TargetOnly	–	2.85±0.63	2.60±0.64	2.61±0.65	2.33±0.50
	ProxyOnly	3.37±0.84	3.19±0.75	3.15±0.86	3.33±1.01	3.02±0.84
	IPW-Transport	2.24±1.10	2.12±0.96	2.26±1.06	2.39±1.15	2.21±0.96
	OM-Transport	2.24±1.10	2.11±0.95	2.27±1.07	2.39±1.16	2.20±0.96
	EntropyBal	2.25±1.10	2.12±0.96	2.27±1.06	2.39±1.15	2.21±0.96
	AnchorOnly	–	2.82±0.67	2.65±0.71	2.64±0.65	2.34±0.51
	Proposed	–	**0.67±0.17**	0.52±0.13	0.42±0.12	0.35±0.16
	Proposed-CF	–	0.86±0.29	**0.50±0.14**	**0.38±0.13**	**0.33±0.17**
	Proposed-B	**1.95±0.92**	2.07±0.92	2.25±1.06	2.30±1.20	2.14±0.98
50	TargetOnly	–	5.22±0.93	4.75±0.86	4.63±0.88	4.57±0.80
	ProxyOnly	5.78±1.22	5.40±0.97	5.30±1.00	5.29±1.19	5.29±1.04
	IPW-Transport	3.26±1.83	3.26±1.53	2.98±1.46	3.14±1.61	3.17±1.52
	OM-Transport	3.25±1.82	3.25±1.54	2.95±1.47	3.07±1.62	3.09±1.53
	EntropyBal	3.40±1.79	3.35±1.49	3.05±1.44	3.19±1.60	3.18±1.52
	AnchorOnly	–	5.20±0.98	4.80±0.90	4.69±0.93	4.63±0.82
	Proposed	–	**1.42±0.65**	**0.73±0.16**	0.57±0.11	0.46±0.12
	Proposed-CF	–	2.67±1.24	1.19±0.43	**0.57±0.11**	**0.40±0.12**
	Proposed-B	**3.25±1.82**	3.07±1.43	2.53±1.24	2.83±1.52	3.03±1.55
100	TargetOnly	–	7.93±1.47	7.57±1.58	7.23±1.31	7.08±1.03
	ProxyOnly	8.21±1.23	8.03±1.48	8.00±1.70	7.81±1.55	7.73±1.30
	IPW-Transport	4.10±2.05	4.13±2.23	4.32±2.43	4.54±2.42	4.61±1.94
	OM-Transport	4.06±2.06	4.02±2.26	4.21±2.45	4.28±2.49	4.28±1.98
	EntropyBal	4.77±1.93	4.74±2.17	4.67±2.39	4.87±2.33	4.81±1.94
	AnchorOnly	–	7.93±1.56	7.64±1.67	7.34±1.39	7.18±1.06
	Proposed	–	**3.10±1.81**	**1.71±0.75**	**0.84±0.20**	0.58±0.11
	Proposed-CF	–	5.32±2.04	3.22±1.66	1.28±0.40	**0.52±0.11**
	Proposed-B	**4.05±2.06**	4.02±2.26	4.20±2.45	3.56±2.10	3.96±2.14

TABLE IX
ATE (MEAN±SD) ACROSS BUDGET AND p. LOWER IS BETTER.

p	Method	(50,0)	(100,50)	(150,100)	(250,200)	(550,500)
10	TargetOnly	–	0.19±0.15	0.14±0.10	0.10±0.07	0.07±0.06
	ProxyOnly	0.69±0.52	0.56±0.45	0.53±0.48	0.69±0.48	0.67±0.48
	IPW-Transport	0.77±0.56	0.74±0.50	0.52±0.42	0.64±0.50	0.70±0.57
	OM-Transport	0.77±0.56	0.74±0.50	0.52±0.42	0.64±0.49	0.70±0.57
	EntropyBal	0.77±0.56	0.74±0.51	0.52±0.42	0.64±0.50	0.70±0.57
	AnchorOnly	–	0.21±0.16	0.13±0.11	0.10±0.09	0.08±0.06
	Proposed	–	**0.08±0.06**	**0.05±0.04**	0.04±0.03	**0.03±0.02**
	Proposed-CF	–	0.10±0.07	0.06±0.05	**0.04±0.03**	0.03±0.02
	Proposed-B	0.72±0.54	0.73±0.52	0.50±0.39	0.64±0.48	0.71±0.57
20	TargetOnly	–	0.29±0.26	0.20±0.18	0.17±0.13	0.10±0.06
	ProxyOnly	0.85±0.66	0.71±0.57	0.74±0.53	0.81±0.69	0.78±0.62
	IPW-Transport	0.80±0.60	0.66±0.56	0.74±0.57	0.75±0.68	0.73±0.51
	OM-Transport	0.80±0.60	0.65±0.55	0.75±0.56	0.74±0.69	0.73±0.51
	EntropyBal	0.79±0.59	0.66±0.56	0.74±0.57	0.75±0.68	0.73±0.51
	AnchorOnly	–	0.29±0.22	0.22±0.17	0.17±0.13	0.10±0.07
	Proposed	–	**0.10±0.07**	**0.06±0.04**	0.05±0.04	**0.03±0.02**
	Proposed-CF	–	0.12±0.11	0.08±0.06	**0.05±0.04**	0.03±0.02
	Proposed-B	0.73±0.56	0.69±0.54	0.78±0.51	0.76±0.68	0.72±0.53
50	TargetOnly	–	0.53±0.45	0.39±0.32	0.29±0.23	0.19±0.15
	ProxyOnly	1.33±1.04	0.94±0.84	1.14±0.89	1.12±0.88	1.12±0.87
	IPW-Transport	0.76±0.59	0.99±0.91	0.85±0.68	0.89±0.74	0.92±0.64

TABLE IX

TABLE IX (CONTINUED)

p	Method	(50,0)	(100,50)	(150,100)	(250,200)	(550,500)
	OM-Transport	**0.74±0.55**	1.01±0.92	0.85±0.67	0.89±0.73	0.92±0.65
	EntropyBal	0.76±0.58	0.99±0.92	0.85±0.67	0.89±0.74	0.92±0.64
	AnchorOnly	–	0.55±0.45	0.39±0.32	0.27±0.22	0.17±0.14
	Proposed	–	**0.18±0.19**	**0.08±0.06**	**0.05±0.04**	**0.03±0.02**
	Proposed-CF	–	0.34±0.31	0.13±0.13	0.06±0.04	0.03±0.02
	Proposed-B	0.74±0.54	0.92±0.83	0.81±0.58	0.83±0.73	0.92±0.66
100	TargetOnly	–	0.77±0.61	0.57±0.45	0.45±0.31	0.31±0.23
	ProxyOnly	1.86±1.60	1.46±1.16	1.35±1.10	1.49±1.00	1.45±1.15
	IPW-Transport	1.00±0.89	0.98±0.71	0.97±0.97	1.14±1.02	1.26±1.06
	OM-Transport	0.99±0.87	0.93±0.71	0.95±0.99	1.08±1.00	1.21±1.01
	EntropyBal	**0.98±0.87**	1.02±0.76	0.98±0.95	1.15±1.02	1.27±1.08
	AnchorOnly	–	0.71±0.62	0.61±0.51	0.44±0.34	0.30±0.23
	Proposed	–	**0.36±0.32**	**0.15±0.14**	**0.07±0.05**	0.04±0.03
	Proposed-CF	–	0.52±0.43	0.28±0.25	0.11±0.10	**0.03±0.02**
	Proposed-B	0.99±0.87	0.93±0.71	0.95±0.99	0.98±0.83	1.12±0.97

TABLE X

SPEARMAN (MEAN±SD) ACROSS BUDGET AND p. HIGHER IS BETTER.

p	Method	(50,0)	(100,50)	(150,100)	(250,200)	(550,500)
10	TargetOnly	–	0.80±0.06	0.85±0.04	0.86±0.03	0.88±0.03
	ProxyOnly	0.64±0.15	0.68±0.16	0.72±0.15	0.71±0.14	0.71±0.20
	IPW-Transport	0.81±0.17	0.77±0.21	0.80±0.19	0.78±0.19	0.78±0.25
	OM-Transport	0.81±0.17	0.77±0.21	0.80±0.19	0.78±0.19	0.78±0.25
	EntropyBal	0.81±0.17	0.77±0.21	0.80±0.19	0.78±0.19	0.78±0.25
	AnchorOnly	–	0.80±0.07	0.85±0.05	0.86±0.04	0.87±0.03
	Proposed	–	**0.98±0.02**	0.98±0.02	0.99±0.01	0.99±0.01
	Proposed-CF	–	0.97±0.02	**0.98±0.02**	**0.99±0.01**	**0.99±0.01**
	Proposed-B	**0.85±0.14**	0.79±0.21	0.82±0.17	0.80±0.19	0.78±0.25
20	TargetOnly	–	0.69±0.08	0.75±0.05	0.78±0.04	0.79±0.03
	ProxyOnly	0.51±0.13	0.60±0.12	0.61±0.12	0.62±0.13	0.62±0.15
	IPW-Transport	0.81±0.16	0.83±0.13	0.81±0.16	0.81±0.15	0.81±0.15
	OM-Transport	0.81±0.16	0.83±0.13	0.81±0.16	0.81±0.15	0.81±0.15
	EntropyBal	0.81±0.16	0.83±0.14	0.81±0.16	0.81±0.15	0.81±0.15
	AnchorOnly	–	0.70±0.07	0.75±0.05	0.78±0.04	0.79±0.03
	Proposed	–	**0.98±0.01**	**0.99±0.01**	0.99±0.00	0.99±0.01
	Proposed-CF	–	0.97±0.03	0.99±0.01	**0.99±0.00**	**0.99±0.01**
	Proposed-B	**0.86±0.13**	0.84±0.14	0.81±0.16	0.82±0.16	0.82±0.15
50	TargetOnly	–	0.48±0.08	0.60±0.06	0.65±0.05	0.68±0.04
	ProxyOnly	0.33±0.11	0.43±0.11	0.47±0.08	0.50±0.11	0.51±0.09
	IPW-Transport	0.82±0.16	0.83±0.14	0.85±0.13	0.84±0.15	0.84±0.14
	OM-Transport	0.82±0.16	0.83±0.14	0.85±0.13	0.84±0.15	0.85±0.13
	EntropyBal	0.81±0.16	0.82±0.14	0.84±0.13	0.83±0.15	0.84±0.14
	AnchorOnly	–	0.50±0.08	0.60±0.06	0.64±0.05	0.67±0.04
	Proposed	–	**0.96±0.03**	**0.99±0.00**	**0.99±0.00**	1.00±0.00
	Proposed-CF	–	0.88±0.09	0.98±0.02	0.99±0.00	**1.00±0.00**
	Proposed-B	**0.82±0.16**	0.85±0.13	0.89±0.09	0.86±0.15	0.85±0.13
100	TargetOnly	–	0.34±0.08	0.44±0.06	0.53±0.06	0.58±0.05
	ProxyOnly	0.20±0.09	0.33±0.08	0.34±0.08	0.40±0.08	0.43±0.08
	IPW-Transport	0.84±0.14	0.85±0.12	0.85±0.12	0.83±0.14	0.83±0.11
	OM-Transport	0.85±0.14	0.86±0.12	0.85±0.12	0.84±0.14	0.85±0.11
	EntropyBal	0.80±0.14	0.82±0.13	0.82±0.13	0.81±0.14	0.82±0.12
	AnchorOnly	–	0.34±0.09	0.44±0.07	0.51±0.06	0.56±0.05
	Proposed	–	**0.91±0.08**	**0.98±0.02**	**0.99±0.00**	1.00±0.00
	Proposed-CF	–	0.76±0.14	0.92±0.06	0.99±0.01	**1.00±0.00**
	Proposed-B	**0.85±0.14**	0.86±0.12	0.85±0.12	0.89±0.11	0.87±0.12

TABLE XI

REGRET (MEAN±SD) ACROSS BUDGET AND p. LOWER IS BETTER.

p	Method	(50,0)	(100,50)	(150,100)	(250,200)	(550,500)
10	TargetOnly	–	0.20±0.09	0.14±0.05	0.13±0.05	0.12±0.05
	ProxyOnly	0.43±0.25	0.36±0.22	0.32±0.22	0.35±0.21	0.35±0.27
	IPW-Transport	0.27±0.25	0.28±0.24	0.22±0.20	0.27±0.24	0.29±0.37
	OM-Transport	0.27±0.25	0.28±0.24	0.22±0.20	0.27±0.25	0.29±0.37
	EntropyBal	0.27±0.24	0.28±0.25	0.22±0.20	0.27±0.24	0.29±0.37
	AnchorOnly	–	0.19±0.08	0.14±0.06	0.13±0.06	0.12±0.06

TABLE XI

TABLE XI (CONTINUED)

p	Method	(50,0)	(100,50)	(150,100)	(250,200)	(550,500)
	Proposed	–	**0.02±0.01**	0.01±0.01	0.01±0.01	0.01±0.01
	Proposed-CF	–	0.02±0.01	**0.01±0.01**	**0.01±0.01**	**0.01±0.01**
	Proposed-B	0.21±0.20	0.26±0.23	0.19±0.18	0.25±0.24	0.29±0.37
20	TargetOnly	–	0.46±0.16	0.35±0.12	0.32±0.11	0.28±0.08
	ProxyOnly	0.75±0.32	0.63±0.29	0.62±0.32	0.67±0.35	0.60±0.37
	IPW-Transport	0.32±0.32	0.28±0.25	0.32±0.29	0.34±0.32	0.31±0.29
	OM-Transport	0.32±0.32	0.28±0.25	0.32±0.29	0.34±0.32	0.31±0.28
	EntropyBal	0.32±0.32	0.28±0.25	0.32±0.28	0.34±0.32	0.31±0.29
	AnchorOnly	–	0.42±0.15	0.36±0.14	0.33±0.10	0.29±0.09
	Proposed	–	**0.02±0.01**	0.01±0.01	0.01±0.01	0.01±0.01
	Proposed-CF	–	0.04±0.03	**0.01±0.01**	**0.01±0.01**	**0.01±0.01**
	Proposed-B	**0.23±0.22**	0.26±0.24	0.31±0.29	0.32±0.33	0.29±0.29
50	TargetOnly	–	1.14±0.30	0.89±0.25	0.77±0.17	0.72±0.15
	ProxyOnly	1.63±0.57	1.34±0.45	1.31±0.46	1.22±0.47	1.20±0.43
	IPW-Transport	0.46±0.53	0.44±0.41	0.37±0.35	0.40±0.40	0.40±0.38
	OM-Transport	**0.45±0.53**	0.45±0.42	0.36±0.34	0.39±0.40	0.39±0.37
	EntropyBal	0.48±0.54	0.46±0.41	0.38±0.35	0.41±0.40	0.40±0.38
	AnchorOnly	–	1.10±0.32	0.92±0.28	0.80±0.21	0.76±0.16
	Proposed	–	**0.08±0.08**	**0.02±0.01**	**0.01±0.01**	0.01±0.00
	Proposed-CF	–	0.27±0.25	0.05±0.04	0.01±0.01	**0.01±0.00**
	Proposed-B	0.45±0.53	0.39±0.38	0.26±0.24	0.33±0.38	0.38±0.38
100	TargetOnly	–	2.04±0.48	1.74±0.44	1.52±0.38	1.36±0.29
	ProxyOnly	2.48±0.63	2.34±0.75	2.19±0.71	2.12±0.74	2.01±0.65
	IPW-Transport	0.52±0.53	0.52±0.55	0.55±0.60	0.62±0.67	0.59±0.49
	OM-Transport	0.51±0.53	0.51±0.54	0.54±0.60	0.58±0.67	0.53±0.47
	EntropyBal	0.64±0.56	0.63±0.58	0.63±0.65	0.68±0.68	0.64±0.52
	AnchorOnly	–	2.02±0.54	1.77±0.49	1.57±0.38	1.43±0.31
	Proposed	–	**0.28±0.30**	**0.07±0.06**	**0.02±0.01**	0.01±0.00
	Proposed-CF	–	0.81±0.61	0.27±0.24	0.04±0.02	**0.01±0.00**
	Proposed-B	**0.51±0.53**	0.51±0.54	0.53±0.60	0.38±0.51	0.47±0.49

B. Source Site Scaling

TABLE XII

PEHE (MEAN±SD) ACROSS SOURCES C AND BUDGET. LOWER IS BETTER.

C	Method	(50,0)	(100,50)	(150,100)	(250,200)	(550,500)
2	TargetOnly	–	5.33±0.92	4.95±1.18	4.70±0.75	4.46±0.91
	ProxyOnly	5.67±1.05	5.62±1.06	5.55±1.37	5.42±1.16	5.19±1.22
	IPW-Transport	3.64±1.45	4.17±1.88	3.78±1.85	4.03±1.94	3.74±1.97
	OM-Transport	3.54±1.45	4.01±1.93	3.59±1.89	3.86±1.93	3.52±1.98
	EntropyBal	3.76±1.45	4.23±1.87	3.79±1.85	4.04±1.94	3.73±1.98
	AnchorOnly	–	5.39±1.00	5.01±1.26	4.75±0.78	4.52±0.94
	Proposed	–	**1.76±0.72**	**0.84±0.16**	0.62±0.12	0.45±0.11
	Proposed-CF	–	3.17±1.25	1.35±0.54	**0.60±0.12**	**0.39±0.12**
	Proposed-B	**3.51±1.47**	3.89±1.95	3.51±1.89	3.78±1.89	3.48±2.00
5	TargetOnly	–	5.12±0.95	4.79±0.96	4.45±0.70	4.56±0.86
	ProxyOnly	5.57±0.87	5.35±0.92	5.25±1.12	5.07±0.91	5.21±1.09
	IPW-Transport	3.33±1.45	3.27±1.40	3.43±1.52	3.18±1.48	3.59±1.74
	OM-Transport	3.24±1.46	3.12±1.44	3.25±1.58	2.97±1.51	3.35±1.75
	EntropyBal	3.53±1.47	3.36±1.38	3.48±1.52	3.19±1.48	3.56±1.74
	AnchorOnly	–	5.17±1.03	4.85±1.04	4.50±0.76	4.61±0.87
	Proposed	–	**1.52±0.70**	**0.78±0.16**	0.60±0.12	0.45±0.12
	Proposed-CF	–	2.76±1.11	1.16±0.34	**0.58±0.12**	**0.39±0.13**
	Proposed-B	**3.24±1.46**	3.01±1.37	2.81±1.43	2.70±1.54	3.26±1.77
10	TargetOnly	–	5.53±1.20	4.96±0.81	4.73±0.78	4.66±0.93
	ProxyOnly	5.64±0.90	5.75±1.30	5.53±0.94	5.41±1.06	5.41±1.37
	IPW-Transport	3.14±1.29	3.49±1.77	3.39±1.39	3.41±1.71	3.46±1.80
	OM-Transport	3.09±1.30	3.42±1.75	3.28±1.42	3.30±1.71	3.33±1.87
	EntropyBal	3.31±1.24	3.59±1.74	3.47±1.38	3.43±1.71	3.45±1.81
	AnchorOnly	–	5.53±1.30	5.04±0.85	4.79±0.81	4.71±0.94
	Proposed	–	**1.63±0.84**	**0.73±0.18**	0.59±0.12	0.45±0.14
	Proposed-CF	–	2.93±1.42	1.19±0.39	**0.57±0.13**	**0.39±0.15**
	Proposed-B	**3.09±1.30**	3.14±1.62	2.78±1.31	2.96±1.71	3.18±1.85
20	TargetOnly	–	5.40±1.01	4.78±0.77	4.69±0.86	4.47±0.83
	ProxyOnly	5.71±0.96	5.61±1.06	5.29±0.97	5.30±1.18	5.12±1.00
	IPW-Transport	2.86±1.19	3.14±1.42	2.88±1.38	3.16±1.51	3.02±1.56
	OM-Transport	**2.84±1.19**	3.13±1.42	2.83±1.39	3.07±1.55	2.91±1.58
	EntropyBal	2.99±1.16	3.23±1.41	2.96±1.36	3.22±1.49	3.03±1.55
	AnchorOnly	–	5.39±1.04	4.84±0.81	4.75±0.91	4.52±0.83

TABLE XII

TABLE XII (CONTINUED)

C	Method	(50,0)	(100,50)	(150,100)	(250,200)	(550,500)
	Proposed	–	**1.42±0.61**	**0.69±0.16**	0.58±0.14	0.44±0.10
	Proposed-CF	–	2.78±1.36	1.08±0.31	**0.57±0.14**	**0.39±0.11**
	Proposed-B	2.84±1.19	2.92±1.28	2.39±1.21	2.72±1.55	2.63±1.53
50	TargetOnly	–	5.36±1.19	5.05±1.18	4.70±0.89	4.56±0.77
	ProxyOnly	5.55±0.85	5.54±1.28	5.62±1.50	5.31±1.09	5.25±1.02
	IPW-Transport	2.74±1.17	3.11±1.77	3.23±1.97	3.03±1.48	2.89±1.48
	OM-Transport	2.74±1.17	3.11±1.77	3.22±1.97	3.03±1.50	2.85±1.50
	EntropyBal	2.84±1.16	3.16±1.74	3.28±1.94	3.08±1.46	2.92±1.48
	AnchorOnly	–	5.34±1.21	5.10±1.24	4.74±0.92	4.62±0.78
	Proposed	–	**1.50±0.80**	**0.66±0.15**	0.54±0.11	0.45±0.16
	Proposed-CF	–	2.70±1.40	1.26±0.57	**0.54±0.12**	**0.41±0.16**
	Proposed-B	**2.74±1.17**	2.91±1.67	2.24±1.61	2.28±1.44	2.38±1.51

TABLE XIII

ATE (MEAN±SD) ACROSS SOURCES C AND BUDGET. LOWER IS BETTER.

C	Method	(50,0)	(100,50)	(150,100)	(250,200)	(550,500)
2	TargetOnly	–	0.49±0.41	0.40±0.35	0.31±0.24	0.21±0.15
	ProxyOnly	1.29±1.05	1.16±0.97	1.08±0.96	1.20±1.20	1.13±0.94
	IPW-Transport	0.79±0.63	1.13±0.95	0.88±0.64	1.12±0.97	0.79±0.66
	OM-Transport	0.79±0.64	1.11±0.93	0.88±0.66	1.12±0.97	0.80±0.67
	EntropyBal	0.82±0.63	1.14±0.96	0.88±0.64	1.13±0.97	0.79±0.66
	AnchorOnly	–	0.54±0.46	0.38±0.33	0.30±0.23	0.21±0.14
	Proposed	–	**0.21±0.17**	**0.08±0.07**	**0.05±0.03**	**0.03±0.03**
	Proposed-CF	–	0.40±0.44	0.15±0.14	0.06±0.04	0.04±0.03
	Proposed-B	**0.79±0.64**	1.09±0.92	0.91±0.70	1.13±0.88	0.78±0.67
5	TargetOnly	–	0.49±0.40	0.36±0.28	0.27±0.19	0.23±0.16
	ProxyOnly	1.29±0.92	1.00±0.72	0.92±0.79	1.05±0.87	1.13±0.77
	IPW-Transport	0.91±0.70	0.90±0.74	0.94±0.77	0.90±0.85	1.02±1.04
	OM-Transport	0.89±0.67	0.89±0.73	0.91±0.74	0.87±0.86	1.02±0.99
	EntropyBal	0.90±0.69	0.90±0.73	0.94±0.77	0.89±0.85	1.02±1.04
	AnchorOnly	–	0.53±0.36	0.36±0.29	0.24±0.19	0.21±0.15
	Proposed	–	**0.18±0.15**	**0.08±0.06**	0.05±0.04	0.04±0.03
	Proposed-CF	–	0.37±0.36	0.15±0.12	**0.05±0.04**	**0.04±0.03**
	Proposed-B	0.89±0.66	0.87±0.70	0.82±0.66	0.84±0.81	1.02±1.00
10	TargetOnly	–	0.59±0.42	0.36±0.27	0.26±0.22	0.21±0.17
	ProxyOnly	1.33±1.09	1.16±0.92	1.13±0.91	1.03±0.82	1.19±1.02
	IPW-Transport	**0.86±0.65**	1.00±0.83	1.05±0.77	1.01±0.87	0.96±0.75
	OM-Transport	0.86±0.63	1.01±0.82	1.02±0.74	1.00±0.81	0.99±0.78
	EntropyBal	0.87±0.64	0.99±0.82	1.05±0.77	1.01±0.86	0.97±0.76
	AnchorOnly	–	0.55±0.41	0.34±0.29	0.27±0.20	0.22±0.19
	Proposed	–	**0.24±0.25**	**0.08±0.07**	**0.05±0.03**	**0.03±0.02**
	Proposed-CF	–	0.35±0.37	0.13±0.10	0.06±0.04	0.03±0.02
	Proposed-B	0.86±0.63	0.91±0.80	0.89±0.63	0.95±0.79	0.92±0.79
20	TargetOnly	–	0.53±0.41	0.37±0.31	0.26±0.22	0.16±0.13
	ProxyOnly	1.32±0.99	1.05±0.84	1.04±0.84	1.08±1.00	1.09±0.78
	IPW-Transport	0.83±0.64	0.81±0.63	0.76±0.60	0.99±0.83	0.80±0.59
	OM-Transport	**0.82±0.63**	0.82±0.65	0.75±0.57	0.98±0.84	0.80±0.57
	EntropyBal	0.82±0.63	0.80±0.64	0.76±0.59	0.99±0.83	0.80±0.58
	AnchorOnly	–	0.58±0.43	0.37±0.28	0.27±0.21	0.15±0.12
	Proposed	–	**0.15±0.13**	**0.08±0.07**	**0.05±0.04**	**0.03±0.02**
	Proposed-CF	–	0.29±0.44	0.13±0.10	0.05±0.04	0.03±0.03
	Proposed-B	0.82±0.63	0.77±0.59	0.68±0.50	0.91±0.85	0.75±0.58
50	TargetOnly	–	0.57±0.44	0.39±0.31	0.27±0.23	0.21±0.16
	ProxyOnly	1.25±0.89	1.01±0.89	1.16±1.01	1.02±0.84	1.17±0.92
	IPW-Transport	0.78±0.58	0.86±0.78	0.88±0.88	0.81±0.59	0.82±0.63
	OM-Transport	0.77±0.58	0.86±0.78	0.87±0.87	0.81±0.58	0.80±0.62
	EntropyBal	0.78±0.57	0.86±0.77	0.88±0.87	0.81±0.59	0.82±0.63
	AnchorOnly	–	0.54±0.43	0.38±0.29	0.27±0.21	0.20±0.15
	Proposed	–	**0.18±0.14**	**0.08±0.07**	**0.05±0.04**	**0.03±0.02**
	Proposed-CF	–	0.29±0.28	0.14±0.12	0.06±0.04	0.03±0.03
	Proposed-B	**0.77±0.58**	0.80±0.74	0.69±0.67	0.74±0.56	0.72±0.60

TABLE XIV
SPEARMAN (MEAN±SD) ACROSS SOURCES C AND BUDGET. HIGHER IS BETTER.

C	Method	(50,0)	(100,50)	(150,100)	(250,200)	(550,500)
2	TargetOnly	–	0.47±0.08	0.60±0.06	0.66±0.05	0.68±0.04
	ProxyOnly	0.34±0.09	0.41±0.10	0.45±0.10	0.49±0.10	0.49±0.12
	IPW-Transport	0.78±0.15	0.74±0.19	0.78±0.16	0.75±0.19	0.77±0.20
	OM-Transport	0.79±0.15	0.75±0.19	0.80±0.17	0.77±0.19	0.79±0.20
	EntropyBal	0.77±0.15	0.73±0.19	0.78±0.16	0.75±0.19	0.77±0.20
	AnchorOnly	–	0.46±0.08	0.59±0.06	0.65±0.04	0.67±0.05
	Proposed	–	**0.94±0.05**	**0.99±0.00**	**0.99±0.00**	1.00±0.00
	Proposed-CF	–	0.82±0.14	0.97±0.02	0.99±0.00	**1.00±0.00**
	Proposed-B	**0.80±0.15**	0.76±0.19	0.81±0.17	0.78±0.19	0.79±0.21
5	TargetOnly	–	0.49±0.09	0.59±0.06	0.66±0.05	0.67±0.04
	ProxyOnly	0.33±0.10	0.43±0.11	0.46±0.09	0.50±0.09	0.52±0.09
	IPW-Transport	0.81±0.15	0.82±0.13	0.81±0.14	0.83±0.13	0.80±0.15
	OM-Transport	0.82±0.15	0.84±0.13	0.82±0.14	0.85±0.13	0.82±0.15
	EntropyBal	0.79±0.15	0.82±0.13	0.80±0.14	0.83±0.13	0.80±0.15
	AnchorOnly	–	0.49±0.09	0.59±0.07	0.65±0.06	0.66±0.04
	Proposed	–	**0.96±0.04**	**0.99±0.01**	**0.99±0.00**	1.00±0.00
	Proposed-CF	–	0.86±0.09	0.98±0.01	0.99±0.00	**1.00±0.00**
	Proposed-B	**0.82±0.15**	0.85±0.12	0.87±0.11	0.87±0.13	0.83±0.15
10	TargetOnly	–	0.47±0.08	0.59±0.06	0.66±0.05	0.67±0.04
	ProxyOnly	0.33±0.11	0.43±0.10	0.47±0.09	0.49±0.10	0.51±0.11
	IPW-Transport	0.83±0.11	0.82±0.14	0.82±0.12	0.82±0.14	0.82±0.14
	OM-Transport	0.84±0.11	0.83±0.13	0.83±0.12	0.83±0.14	0.83±0.15
	EntropyBal	0.82±0.11	0.81±0.14	0.82±0.13	0.82±0.14	0.82±0.14
	AnchorOnly	–	0.48±0.09	0.59±0.07	0.65±0.05	0.66±0.04
	Proposed	–	**0.96±0.04**	**0.99±0.00**	**0.99±0.00**	1.00±0.00
	Proposed-CF	–	0.87±0.10	0.98±0.01	0.99±0.00	**1.00±0.00**
	Proposed-B	**0.84±0.11**	0.86±0.12	0.88±0.12	0.86±0.14	0.84±0.15
20	TargetOnly	–	0.47±0.08	0.59±0.06	0.65±0.04	0.68±0.04
	ProxyOnly	0.32±0.11	0.42±0.11	0.45±0.09	0.50±0.09	0.52±0.10
	IPW-Transport	0.87±0.09	0.84±0.12	0.86±0.11	0.84±0.12	0.84±0.13
	OM-Transport	0.87±0.09	0.84±0.12	0.86±0.11	0.85±0.12	0.85±0.13
	EntropyBal	0.86±0.09	0.83±0.12	0.85±0.11	0.84±0.11	0.84±0.13
	AnchorOnly	–	0.49±0.09	0.58±0.07	0.64±0.05	0.67±0.04
	Proposed	–	**0.97±0.03**	**0.99±0.00**	**0.99±0.00**	1.00±0.00
	Proposed-CF	–	0.88±0.08	0.98±0.01	0.99±0.00	**1.00±0.00**
	Proposed-B	**0.87±0.09**	0.87±0.10	0.90±0.09	0.88±0.11	0.88±0.13
50	TargetOnly	–	0.47±0.09	0.58±0.06	0.65±0.05	0.68±0.04
	ProxyOnly	0.33±0.10	0.43±0.11	0.45±0.12	0.49±0.10	0.52±0.12
	IPW-Transport	0.87±0.09	0.84±0.14	0.84±0.13	0.85±0.11	0.86±0.13
	OM-Transport	0.87±0.09	0.84±0.14	0.84±0.13	0.85±0.12	0.86±0.13
	EntropyBal	0.86±0.09	0.84±0.14	0.84±0.13	0.84±0.11	0.86±0.13
	AnchorOnly	–	0.49±0.09	0.58±0.07	0.64±0.05	0.67±0.04
	Proposed	–	**0.96±0.04**	**0.99±0.00**	**1.00±0.00**	1.00±0.00
	Proposed-CF	–	0.89±0.08	0.98±0.01	0.99±0.00	**1.00±0.00**
	Proposed-B	**0.87±0.09**	0.87±0.12	0.92±0.09	0.91±0.11	0.90±0.14

TABLE XV
REGRET (MEAN±SD) ACROSS SOURCES C AND BUDGET. LOWER IS BETTER.

C	Method	(50,0)	(100,50)	(150,100)	(250,200)	(550,500)
2	TargetOnly	–	1.18±0.24	0.92±0.28	0.79±0.18	0.71±0.19
	ProxyOnly	1.58±0.51	1.50±0.45	1.39±0.56	1.34±0.60	1.23±0.50
	IPW-Transport	0.50±0.42	0.67±0.55	0.55±0.56	0.63±0.56	0.56±0.57
	OM-Transport	0.48±0.41	0.63±0.55	0.51±0.55	0.60±0.57	0.52±0.59
	EntropyBal	0.52±0.42	0.68±0.54	0.55±0.56	0.63±0.56	0.55±0.57
	AnchorOnly	–	1.20±0.30	0.93±0.28	0.82±0.17	0.75±0.21
	Proposed	–	**0.12±0.10**	**0.02±0.01**	**0.01±0.01**	0.01±0.00
	Proposed-CF	–	0.40±0.38	0.06±0.05	0.01±0.01	**0.01±0.00**
	Proposed-B	**0.48±0.41**	0.61±0.56	0.49±0.55	0.58±0.54	0.52±0.59
5	TargetOnly	–	1.12±0.31	0.91±0.25	0.76±0.20	0.73±0.19
	ProxyOnly	1.52±0.40	1.35±0.41	1.28±0.49	1.21±0.40	1.17±0.39
	IPW-Transport	0.45±0.39	0.43±0.36	0.46±0.41	0.42±0.38	0.52±0.55
	OM-Transport	**0.44±0.40**	0.40±0.36	0.43±0.42	0.38±0.37	0.47±0.52
	EntropyBal	0.50±0.41	0.44±0.37	0.47±0.41	0.42±0.38	0.52±0.55
	AnchorOnly	–	1.11±0.34	0.91±0.26	0.79±0.22	0.76±0.19
	Proposed	–	**0.09±0.09**	**0.02±0.01**	**0.01±0.01**	0.01±0.01
	Proposed-CF	–	0.30±0.23	0.05±0.02	0.01±0.01	**0.01±0.01**
	Proposed-B	0.44±0.40	0.37±0.34	0.32±0.36	0.33±0.37	0.46±0.53
10	TargetOnly	–	1.21±0.40	0.91±0.22	0.79±0.18	0.75±0.18
	ProxyOnly	1.53±0.45	1.43±0.60	1.32±0.41	1.28±0.44	1.25±0.59

TABLE XV
TABLE XV (CONTINUED)

C	Method	(50,0)	(100,50)	(150,100)	(250,200)	(550,500)
	IPW-Transport	0.39±0.31	0.48±0.48	0.45±0.34	0.49±0.50	0.49±0.50
	OM-Transport	0.38±0.31	0.46±0.46	0.42±0.34	0.46±0.48	0.46±0.51
	EntropyBal	0.42±0.31	0.50±0.50	0.47±0.35	0.49±0.50	0.49±0.50
	AnchorOnly	–	1.17±0.41	0.93±0.24	0.82±0.19	0.77±0.18
	Proposed	–	**0.10±0.10**	**0.02±0.01**	**0.01±0.01**	0.01±0.01
	Proposed-CF	–	0.32±0.32	0.05±0.03	0.01±0.01	**0.01±0.01**
	Proposed-B	**0.38±0.31**	0.39±0.41	0.31±0.30	0.38±0.48	0.43±0.51
20	TargetOnly	–	1.20±0.30	0.90±0.22	0.81±0.22	0.72±0.17
	ProxyOnly	1.52±0.46	1.43±0.51	1.34±0.46	1.18±0.47	1.16±0.35
	IPW-Transport	0.32±0.26	0.40±0.34	0.35±0.33	0.40±0.34	0.39±0.39
	OM-Transport	0.31±0.26	0.39±0.34	0.34±0.34	0.38±0.34	0.37±0.38
	EntropyBal	0.34±0.26	0.41±0.34	0.36±0.33	0.41±0.33	0.39±0.39
	AnchorOnly	–	1.15±0.33	0.91±0.23	0.84±0.22	0.73±0.17
	Proposed	–	**0.08±0.08**	**0.02±0.01**	**0.01±0.01**	0.01±0.00
	Proposed-CF	–	0.27±0.22	0.04±0.02	0.01±0.01	**0.01±0.00**
	Proposed-B	**0.31±0.26**	0.33±0.28	0.24±0.25	0.31±0.33	0.31±0.37
50	TargetOnly	–	1.21±0.37	0.96±0.28	0.82±0.22	0.72±0.17
	ProxyOnly	1.53±0.39	1.44±0.59	1.43±0.72	1.27±0.45	1.23±0.50
	IPW-Transport	0.31±0.25	0.42±0.47	0.44±0.54	0.38±0.36	0.35±0.43
	OM-Transport	0.31±0.24	0.42±0.48	0.44±0.54	0.38±0.36	0.35±0.42
	EntropyBal	0.33±0.25	0.43±0.46	0.45±0.52	0.39±0.35	0.36±0.43
	AnchorOnly	–	1.16±0.38	0.97±0.30	0.83±0.22	0.74±0.17
	Proposed	–	**0.08±0.09**	**0.01±0.01**	**0.01±0.01**	0.01±0.01
	Proposed-CF	–	0.29±0.34	0.05±0.04	0.01±0.01	**0.01±0.01**
	Proposed-B	**0.31±0.25**	0.36±0.43	0.21±0.42	0.24±0.31	0.26±0.43

C. Sensitivity to A5 Violations

TABLE XVI
PEHE (MEAN±SD) UNDER A5 VIOLATIONS. LOWER IS BETTER.

s/p	Method	$\Gamma = 0.0$	$\Gamma = 0.5$	$\Gamma = 1.0$
0.05	TargetOnly	4.41±0.70	4.25±0.53	4.30±0.57
	ProxyOnly	4.97±0.89	4.72±0.68	4.84±0.72
	IPW-Transport	2.79±1.25	2.13±1.02	2.20±1.11
	OM-Transport	2.67±1.26	2.04±1.03	2.10±1.13
	EntropyBal	2.80±1.25	2.14±1.02	2.21±1.11
	AnchorOnly	4.45±0.70	4.28±0.53	4.34±0.56
	Proposed	0.44±0.11	**1.16±0.68**	**1.92±1.19**
	Proposed-CF	**0.38±0.12**	1.17±0.69	2.04±1.25
	Proposed-B	2.64±1.21	1.92±0.87	2.08±1.13
0.20	TargetOnly	4.80±0.75	4.32±0.55	4.43±0.60
	ProxyOnly	5.62±0.97	4.83±0.68	4.98±0.73
	IPW-Transport	4.18±1.29	2.53±0.89	2.69±1.02
	OM-Transport	4.07±1.31	2.46±0.91	2.56±1.06
	EntropyBal	4.19±1.29	2.54±0.90	2.71±1.01
	AnchorOnly	4.87±0.77	4.37±0.57	4.47±0.62
	Proposed	0.48±0.15	1.42±0.64	**2.41±1.18**
	Proposed-CF	**0.42±0.15**	**1.40±0.65**	2.57±1.22
	Proposed-B	4.03±1.34	2.43±0.95	2.55±1.06
1.00	TargetOnly	4.92±0.92	4.23±0.58	4.60±0.84
	ProxyOnly	5.75±1.21	4.77±0.71	5.04±0.94
	IPW-Transport	4.12±1.68	2.36±0.91	2.82±1.21
	OM-Transport	4.02±1.71	2.28±0.91	2.69±1.24
	EntropyBal	4.14±1.68	2.37±0.91	2.83±1.21
	AnchorOnly	4.99±0.94	4.29±0.60	4.63±0.84
	Proposed	0.45±0.13	1.29±0.64	**2.54±1.35**
	Proposed-CF	**0.39±0.13**	**1.28±0.64**	2.69±1.38
	Proposed-B	3.98±1.74	2.34±1.05	2.67±1.25

TABLE XVII
ATE (MEAN±SD) UNDER A5 VIOLATIONS. LOWER IS BETTER.

s/p	Method	$\Gamma = 0.0$	$\Gamma = 0.5$	$\Gamma = 1.0$
0.05	TargetOnly	0.22±0.14	0.21±0.17	0.16±0.12
	ProxyOnly	0.92±0.75	0.91±0.74	0.98±0.79
	IPW-Transport	0.80±0.60	0.65±0.51	0.61±0.47

TABLE XVII

TABLE XVII (CONTINUED)

s/p	Method	$\Gamma = 0.0$	$\Gamma = 0.5$	$\Gamma = 1.0$
	OM-Transport	0.79±0.58	0.65±0.50	0.60±0.47
	EntropyBal	0.80±0.60	0.65±0.51	0.61±0.46
	AnchorOnly	0.20±0.14	0.19±0.14	0.15±0.12
	Proposed	0.04±0.03	**0.07±0.07**	**0.09±0.07**
	Proposed-CF	**0.04±0.03**	0.07±0.07	0.10±0.08
	Proposed-B	0.83±0.62	0.60±0.44	0.57±0.43
0.20	TargetOnly	0.20±0.15	0.23±0.16	0.18±0.14
	ProxyOnly	1.14±0.76	0.94±0.79	0.98±0.79
	IPW-Transport	1.08±0.74	0.75±0.59	0.64±0.42
	OM-Transport	1.08±0.75	0.74±0.60	0.60±0.41
	EntropyBal	1.07±0.74	0.74±0.59	0.64±0.42
	AnchorOnly	0.21±0.17	0.22±0.15	0.16±0.15
	Proposed	**0.03±0.03**	**0.09±0.06**	0.12±0.11
	Proposed-CF	0.03±0.02	0.09±0.07	**0.11±0.11**
	Proposed-B	1.06±0.74	0.71±0.58	0.54±0.40
1.00	TargetOnly	0.23±0.20	0.18±0.14	0.20±0.16
	ProxyOnly	1.26±0.91	1.10±0.73	0.88±0.77
	IPW-Transport	0.99±0.84	0.76±0.61	0.65±0.49
	OM-Transport	0.97±0.81	0.75±0.60	0.63±0.46
	EntropyBal	0.98±0.84	0.76±0.61	0.64±0.49
	AnchorOnly	0.24±0.22	0.18±0.13	0.20±0.16
	Proposed	**0.04±0.03**	**0.07±0.05**	**0.13±0.14**
	Proposed-CF	0.04±0.03	0.07±0.05	0.14±0.14
	Proposed-B	0.96±0.80	0.77±0.61	0.56±0.43

TABLE XVIII

SPEARMAN (MEAN±SD) UNDER A5 VIOLATIONS. HIGHER IS BETTER.

s/p	Method	$\Gamma = 0.0$	$\Gamma = 0.5$	$\Gamma = 1.0$
0.05	TargetOnly	0.68±0.04	0.67±0.04	0.67±0.04
	ProxyOnly	0.54±0.09	0.55±0.08	0.55±0.08
	IPW-Transport	0.87±0.11	0.91±0.08	0.91±0.08
	OM-Transport	0.88±0.11	0.92±0.08	0.92±0.08
	EntropyBal	0.87±0.11	0.91±0.08	0.91±0.08
	AnchorOnly	0.67±0.04	0.66±0.04	0.66±0.04
	Proposed	1.00±0.00	**0.97±0.03**	**0.92±0.08**
	Proposed-CF	**1.00±0.00**	0.97±0.03	0.91±0.09
	Proposed-B	0.88±0.10	0.93±0.06	0.92±0.08
0.20	TargetOnly	0.68±0.04	0.67±0.04	0.65±0.05
	ProxyOnly	0.48±0.10	0.55±0.07	0.52±0.08
	IPW-Transport	0.75±0.13	0.89±0.07	0.87±0.08
	OM-Transport	0.76±0.13	0.90±0.06	0.88±0.08
	EntropyBal	0.74±0.13	0.89±0.07	0.87±0.08
	AnchorOnly	0.67±0.04	0.66±0.04	0.64±0.05
	Proposed	1.00±0.00	**0.96±0.03**	**0.88±0.09**
	Proposed-CF	**1.00±0.00**	0.96±0.03	0.87±0.09
	Proposed-B	0.76±0.13	0.90±0.07	0.88±0.08
1.00	TargetOnly	0.68±0.04	0.67±0.04	0.64±0.05
	ProxyOnly	0.48±0.11	0.55±0.07	0.53±0.07
	IPW-Transport	0.75±0.17	0.90±0.07	0.86±0.09
	OM-Transport	0.76±0.17	0.91±0.06	0.88±0.09
	EntropyBal	0.75±0.17	0.90±0.07	0.86±0.09
	AnchorOnly	0.67±0.04	0.66±0.05	0.63±0.05
	Proposed	1.00±0.00	**0.97±0.03**	**0.88±0.09**
	Proposed-CF	**1.00±0.00**	0.97±0.03	0.86±0.10
	Proposed-B	0.77±0.17	0.90±0.08	0.88±0.09

TABLE XIX

REGRET (MEAN±SD) UNDER A5 VIOLATIONS. LOWER IS BETTER.

s/p	Method	$\Gamma = 0.0$	$\Gamma = 0.5$	$\Gamma = 1.0$
0.05	TargetOnly	0.71±0.15	0.68±0.14	0.70±0.15
	ProxyOnly	1.11±0.40	1.02±0.33	1.08±0.31
	IPW-Transport	0.32±0.29	0.20±0.19	0.22±0.22
	OM-Transport	0.30±0.28	0.19±0.19	0.20±0.22
	EntropyBal	0.32±0.29	0.20±0.20	0.22±0.22
	AnchorOnly	0.73±0.16	0.71±0.14	0.72±0.15

TABLE XIX
TABLE XIX (CONTINUED)

s/p	Method	$\Gamma = 0.0$	$\Gamma = 0.5$	$\Gamma = 1.0$
	Proposed	0.01±0.00	**0.06±0.07**	**0.18±0.21**
	Proposed-CF	**0.01±0.00**	0.06±0.07	0.20±0.24
	Proposed-B	0.28±0.27	0.16±0.15	0.20±0.22
0.20	TargetOnly	0.78±0.18	0.72±0.14	0.74±0.14
	ProxyOnly	1.32±0.40	1.03±0.29	1.12±0.32
	IPW-Transport	0.64±0.37	0.25±0.17	0.29±0.21
	OM-Transport	0.62±0.37	0.24±0.16	0.27±0.20
	EntropyBal	0.64±0.37	0.25±0.17	0.30±0.21
	AnchorOnly	0.80±0.19	0.75±0.14	0.77±0.16
	Proposed	0.01±0.01	0.08±0.06	**0.25±0.21**
	Proposed-CF	**0.01±0.01**	**0.08±0.06**	0.28±0.24
	Proposed-B	0.61±0.38	0.24±0.17	0.27±0.20
1.00	TargetOnly	0.79±0.19	0.70±0.12	0.79±0.23
	ProxyOnly	1.40±0.52	1.07±0.31	1.14±0.36
	IPW-Transport	0.65±0.50	0.23±0.18	0.32±0.25
	OM-Transport	0.62±0.50	0.22±0.18	0.30±0.25
	EntropyBal	0.65±0.50	0.23±0.18	0.33±0.26
	AnchorOnly	0.83±0.20	0.73±0.14	0.81±0.22
	Proposed	0.01±0.00	**0.07±0.07**	**0.28±0.25**
	Proposed-CF	**0.00±0.00**	0.07±0.07	0.30±0.27
	Proposed-B	0.62±0.50	0.23±0.21	0.30±0.25

D. *CATE Calibration*

The following tables report CATE calibration metrics: Slope (ideal value = 1.0), R^2 (higher is better), and Expected Calibration Error (ECE, lower is better). Best value in each scenario is bolded.

TABLE XX
CATE CALIBRATION: DIM SWEEP AT BUDGET (150,100). SLOPE (IDEAL=1), R^2 (HIGHER=BETTER), ECE (LOWER=BETTER).

Method	p=10			p=20			p=50			p=100		
	Slope	R^2	ECE	Slope	R^2	ECE	Slope	R^2	ECE	Slope	R^2	ECE
TargetOnly	1.18	0.72	0.40	1.38	0.57	0.76	1.56	0.37	1.21	1.45	0.20	1.47
ProxyOnly	1.13	0.54	0.73	1.36	0.40	1.03	1.57	0.24	1.52	1.53	0.13	1.79
IPW-Transport	0.94	0.69	0.65	0.95	0.70	0.87	0.96	0.75	0.99	0.96	0.75	1.18
OM-Transport	0.94	0.69	0.66	0.95	0.70	0.87	0.97	0.75	0.98	0.97	0.76	1.14
EntropyBal	0.94	0.69	0.66	0.94	0.70	0.87	0.95	0.74	0.99	0.91	0.71	1.30
AnchorOnly	1.16	0.71	0.42	1.33	0.55	0.79	1.61	0.36	1.31	1.45	0.19	1.57
Proposed	1.03	**0.97**	0.11	1.04	**0.98**	0.15	1.02	**0.98**	**0.15**	**1.02**	**0.96**	**0.27**
Proposed-CF	**1.00**	**0.97**	**0.09**	**1.00**	**0.98**	**0.11**	0.98	0.96	0.22	0.97	0.85	0.55
Proposed-B	0.95	0.72	0.62	0.94	0.70	0.90	**1.00**	0.82	0.91	0.98	0.76	1.14

TABLE XXI
CATE CALIBRATION: SOURCES SWEEP AT BUDGET (150,100). SLOPE (IDEAL=1), R^2 (HIGHER=BETTER), ECE (LOWER=BETTER).

	C=2			C=5			C=10			C=20			C=50		
Method	Slope	R^2	ECE	Slope	R^2	ECE	Slope	R^2	ECE	Slope	R^2	ECE	Slope	R^2	ECE
TargetOnly	1.48	0.36	1.24	1.50	0.35	1.20	1.48	0.36	1.20	1.54	0.35	1.23	1.50	0.35	1.20
ProxyOnly	1.43	0.22	1.48	1.47	0.23	1.34	1.57	0.24	1.54	1.50	0.22	1.44	1.52	0.22	1.54
IPW-Transport	0.84	0.66	1.32	0.88	0.69	1.18	0.97	0.71	1.14	0.97	0.76	0.86	**0.99**	0.74	0.96
OM-Transport	0.87	0.69	1.25	0.92	0.71	1.09	**0.99**	0.73	1.11	0.98	0.77	0.84	**0.99**	0.74	0.95
EntropyBal	0.84	0.66	1.32	0.87	0.68	1.20	0.95	0.70	1.15	0.95	0.75	0.88	**0.99**	0.73	0.97
AnchorOnly	1.46	0.34	1.27	1.50	0.34	1.27	1.51	0.34	1.31	1.53	0.33	1.26	1.54	0.34	1.31
Proposed	1.03	**0.98**	**0.19**	1.03	**0.98**	**0.17**	1.02	**0.99**	**0.16**	1.02	**0.98**	**0.15**	1.02	**0.99**	**0.15**
Proposed-CF	**0.98**	0.95	0.29	**0.99**	0.96	0.23	0.98	0.96	0.24	0.99	0.96	0.19	0.98	0.96	0.23
Proposed-B	0.88	0.70	1.27	0.94	0.78	0.99	1.01	0.80	1.03	**1.00**	0.83	0.78	1.04	0.87	0.80

TABLE XXII

CATE CALIBRATION UNDER A5 VIOLATIONS. SLOPE (IDEAL=1), R^2 (HIGHER=BETTER), ECE (LOWER=BETTER).

		Γ=0			Γ=0.5			Γ=1		
s/p	Method	Slope	R^2	ECE	Slope	R^2	ECE	Slope	R^2	ECE
0.05	TargetOnly	1.69	0.48	1.33	1.75	0.47	1.32	1.68	0.47	1.26
	ProxyOnly	1.51	0.32	1.37	1.54	0.33	1.36	1.53	0.32	1.45
	IPW-Transport	0.95	0.78	0.92	0.97	0.85	0.71	0.98	0.85	0.64
	OM-Transport	0.97	0.80	0.89	0.98	0.86	0.69	0.99	**0.86**	0.62
	EntropyBal	0.95	0.78	0.93	0.97	0.85	0.71	0.98	0.85	0.64
	AnchorOnly	1.71	0.47	1.34	1.74	0.46	1.30	1.68	0.45	1.25
	Proposed	1.03	0.99	0.14	1.03	**0.95**	0.16	1.04	**0.86**	0.25
	Proposed-CF	**1.00**	**1.00**	**0.06**	**1.00**	0.94	**0.13**	0.99	0.84	**0.20**
	Proposed-B	0.97	0.80	0.93	**1.00**	0.88	0.64	**1.00**	**0.86**	0.60
0.20	TargetOnly	1.77	0.49	1.50	1.69	0.46	1.29	1.67	0.44	1.24
	ProxyOnly	1.43	0.25	1.52	1.56	0.33	1.40	1.46	0.29	1.39
	IPW-Transport	0.90	0.59	1.31	0.97	0.81	0.82	0.98	0.78	0.69
	OM-Transport	0.93	0.61	1.27	0.98	0.82	0.81	0.99	0.80	0.64
	EntropyBal	0.89	0.59	1.31	0.97	0.81	0.82	0.97	0.78	0.69
	AnchorOnly	1.79	0.47	1.56	1.69	0.45	1.28	1.64	0.42	1.22
	Proposed	1.03	0.99	0.15	1.04	0.93	0.22	1.05	0.80	0.30
	Proposed-CF	**1.00**	0.99	**0.06**	**1.00**	0.93	0.15	0.99	0.77	0.27
	Proposed-B	0.93	0.62	1.25	0.97	0.82	0.78	**1.00**	0.80	0.59
1.00	TargetOnly	1.74	0.48	1.55	1.73	0.47	1.29	1.70	0.43	1.32
	ProxyOnly	1.41	0.26	1.61	1.52	0.32	1.44	1.54	0.30	1.36
	IPW-Transport	0.90	0.62	1.29	0.97	0.83	0.82	0.97	0.77	0.71
	OM-Transport	0.93	0.63	1.24	0.99	0.84	0.79	0.99	0.79	0.67
	EntropyBal	0.90	0.61	1.29	0.97	0.83	0.82	0.97	0.77	0.70
	AnchorOnly	1.74	0.46	1.56	1.70	0.45	1.26	1.72	0.42	1.31
	Proposed	1.03	**1.00**	0.14	1.03	0.94	0.20	1.05	0.79	0.31
	Proposed-CF	**1.00**	**1.00**	**0.06**	**1.00**	0.94	0.14	0.99	0.77	0.28
	Proposed-B	0.94	0.63	1.23	0.97	0.83	0.83	**1.00**	0.79	0.61

E. IHDP Semi-Synthetic (Full Metrics)

Full metrics (PEHE, ATE error, Spearman, Regret, ECE) for the IHDP connected and disconnected regimes (mean $\pm$ SD over 50 realizations).

TABLE XXIII
IHDP CONNECTED REGIME: FULL METRICS (PEHE, ATE ERROR, SPEARMAN, POLICY REGRET, ECE).

Method	Budget	PEHE	ATE err	Spearman	Regret	ECE
TargetOnly	(25,25)	4.22 ± 6.07	0.73 ± 1.12	0.58 ± 0.24	0.39 ± 0.81	1.92 ± 2.23
TargetOnly	(25,50)	5.45 ± 7.12	0.91 ± 1.04	0.60 ± 0.20	0.79 ± 1.29	2.09 ± 2.09
TargetOnly	(25,100)	3.09 ± 3.97	0.67 ± 1.11	0.53 ± 0.23	0.36 ± 1.01	1.30 ± 1.34
TargetOnly	(50,25)	3.39 ± 5.17	0.37 ± 0.52	0.71 ± 0.21	0.19 ± 0.34	1.41 ± 1.69
TargetOnly	(50,50)	3.63 ± 5.43	0.41 ± 0.66	0.66 ± 0.22	0.23 ± 0.46	1.23 ± 1.43
TargetOnly	(50,100)	4.58 ± 6.19	0.54 ± 0.76	0.71 ± 0.21	0.34 ± 0.53	1.61 ± 1.95
TargetOnly	(100,25)	2.48 ± 3.33	0.29 ± 0.35	0.80 ± 0.21	0.09 ± 0.14	0.97 ± 1.03
TargetOnly	(100,50)	3.11 ± 4.86	0.32 ± 0.44	0.79 ± 0.19	0.15 ± 0.27	0.98 ± 1.30
TargetOnly	(100,100)	2.97 ± 4.10	0.29 ± 0.45	0.79 ± 0.15	0.15 ± 0.27	**0.91 ± 1.13**
ProxyOnly	(25,25)	3.33 ± 4.71	0.57 ± 0.82	0.64 ± 0.23	0.22 ± 0.39	1.54 ± 1.72
ProxyOnly	(25,50)	4.21 ± 5.59	0.63 ± 0.92	0.72 ± 0.19	0.34 ± 0.52	1.60 ± 2.07
ProxyOnly	(25,100)	2.36 ± 2.75	0.41 ± 0.42	0.67 ± 0.21	0.19 ± 0.42	0.93 ± 0.79
ProxyOnly	(50,25)	3.34 ± 5.30	0.61 ± 0.97	0.69 ± 0.21	0.19 ± 0.43	1.33 ± 1.73
ProxyOnly	(50,50)	3.22 ± 4.70	0.31 ± 0.44	0.69 ± 0.22	0.17 ± 0.30	1.14 ± 1.41
ProxyOnly	(50,100)	3.90 ± 5.21	0.47 ± 0.73	0.73 ± 0.22	0.25 ± 0.42	1.30 ± 1.61
ProxyOnly	(100,25)	3.04 ± 4.11	1.02 ± 1.71	0.78 ± 0.18	0.18 ± 0.31	1.50 ± 1.85
ProxyOnly	(100,50)	3.38 ± 5.22	0.81 ± 1.26	0.78 ± 0.19	0.18 ± 0.31	1.25 ± 1.53
ProxyOnly	(100,100)	3.21 ± 4.38	0.73 ± 1.06	0.78 ± 0.15	0.18 ± 0.31	1.22 ± 1.46
AnchorOnly	(25,25)	3.12 ± 4.32	0.54 ± 0.74	0.65 ± 0.24	0.18 ± 0.36	1.39 ± 1.59
AnchorOnly	(25,50)	3.88 ± 4.87	0.59 ± 0.73	0.72 ± 0.20	0.27 ± 0.39	**1.49 ± 1.59**
AnchorOnly	(25,100)	2.37 ± 2.57	0.46 ± 0.48	0.67 ± 0.22	0.19 ± 0.41	1.01 ± 1.04
AnchorOnly	(50,25)	2.85 ± 4.29	0.28 ± 0.40	0.70 ± 0.22	0.20 ± 0.48	1.06 ± 1.17
AnchorOnly	(50,50)	2.96 ± 4.20	0.25 ± 0.33	0.70 ± 0.22	0.14 ± 0.25	1.08 ± 1.35
AnchorOnly	(50,100)	3.54 ± 4.78	0.35 ± 0.51	0.74 ± 0.23	0.19 ± 0.30	1.20 ± 1.45
AnchorOnly	(100,25)	2.64 ± 3.85	0.22 ± 0.29	0.78 ± 0.19	0.12 ± 0.24	1.06 ± 1.56
AnchorOnly	(100,50)	2.89 ± 4.51	0.22 ± 0.24	0.78 ± 0.19	0.14 ± 0.29	**0.95 ± 1.44**
AnchorOnly	(100,100)	2.77 ± 3.93	**0.24 ± 0.36**	0.79 ± 0.15	0.15 ± 0.32	0.94 ± 1.41
IPW-Transport	(25,25)	3.69 ± 6.23	1.73 ± 3.32	0.81 ± 0.21	0.23 ± 0.53	2.39 ± 4.06
IPW-Transport	(25,50)	3.76 ± 5.55	1.66 ± 3.55	0.84 ± 0.20	0.29 ± 1.00	2.30 ± 3.90
IPW-Transport	(25,100)	2.29 ± 2.21	1.16 ± 1.42	0.81 ± 0.16	0.18 ± 0.38	1.49 ± 1.56
IPW-Transport	(50,25)	3.72 ± 6.55	1.22 ± 2.07	0.78 ± 0.21	0.19 ± 0.46	2.10 ± 3.41
IPW-Transport	(50,50)	3.30 ± 4.75	1.19 ± 1.63	0.77 ± 0.25	0.20 ± 0.40	2.01 ± 2.84
IPW-Transport	(50,100)	4.84 ± 7.40	2.16 ± 4.57	0.80 ± 0.22	0.57 ± 1.36	3.05 ± 5.13
IPW-Transport	(100,25)	3.25 ± 4.53	1.13 ± 1.50	0.79 ± 0.22	0.13 ± 0.19	1.79 ± 2.27
IPW-Transport	(100,50)	3.76 ± 5.95	1.41 ± 2.25	0.79 ± 0.24	0.18 ± 0.55	2.04 ± 3.08
IPW-Transport	(100,100)	3.82 ± 5.09	1.30 ± 1.72	0.80 ± 0.18	0.22 ± 0.49	2.18 ± 2.67
EntropyBal	(25,25)	4.87 ± 8.67	2.45 ± 4.83	0.75 ± 0.20	0.58 ± 1.25	3.19 ± 5.52
EntropyBal	(25,50)	6.41 ± 13.96	3.08 ± 8.81	0.77 ± 0.27	0.62 ± 1.65	4.39 ± 10.99
EntropyBal	(25,100)	3.55 ± 4.33	1.80 ± 2.58	0.77 ± 0.15	0.41 ± 0.75	2.49 ± 3.36
EntropyBal	(50,25)	4.31 ± 6.70	1.54 ± 2.66	0.75 ± 0.22	0.33 ± 0.69	2.64 ± 4.18
EntropyBal	(50,50)	4.07 ± 6.15	1.64 ± 2.50	0.74 ± 0.24	0.40 ± 0.78	2.55 ± 3.46
EntropyBal	(50,100)	9.30 ± 18.58	5.06 ± 11.56	0.74 ± 0.25	1.09 ± 2.13	6.94 ± 14.99
EntropyBal	(100,25)	4.04 ± 5.45	1.49 ± 1.65	0.74 ± 0.24	0.30 ± 0.54	2.52 ± 3.29
EntropyBal	(100,50)	5.35 ± 8.40	1.45 ± 1.52	0.69 ± 0.26	0.50 ± 0.90	3.06 ± 4.59
EntropyBal	(100,100)	4.64 ± 6.75	1.94 ± 3.54	0.75 ± 0.20	0.50 ± 1.14	2.82 ± 4.33
OM-Transport	(25,25)	3.40 ± 5.73	1.43 ± 2.33	0.84 ± 0.20	0.22 ± 0.51	2.22 ± 3.70
OM-Transport	(25,50)	3.46 ± 4.71	1.41 ± 2.61	0.87 ± 0.18	0.23 ± 0.90	2.11 ± 3.08
OM-Transport	(25,100)	2.05 ± 1.98	0.86 ± 0.99	0.85 ± 0.16	0.14 ± 0.25	1.35 ± 1.41
OM-Transport	(50,25)	3.51 ± 6.35	1.10 ± 1.75	0.81 ± 0.20	0.14 ± 0.25	2.00 ± 3.38
OM-Transport	(50,50)	3.10 ± 4.75	1.02 ± 1.21	0.84 ± 0.17	0.13 ± 0.28	1.86 ± 2.67
OM-Transport	(50,100)	3.99 ± 5.71	1.58 ± 2.61	0.86 ± 0.17	0.34 ± 0.88	2.50 ± 3.67
OM-Transport	(100,25)	3.22 ± 4.86	1.14 ± 1.54	0.84 ± 0.18	0.08 ± 0.12	1.86 ± 2.65
OM-Transport	(100,50)	3.63 ± 6.12	1.21 ± 1.74	0.86 ± 0.19	0.14 ± 0.38	2.02 ± 3.24
OM-Transport	(100,100)	3.59 ± 5.13	1.17 ± 1.43	0.85 ± 0.16	0.14 ± 0.28	2.15 ± 2.94
Proposed	(25,25)	**2.46 ± 4.36**	0.53 ± 1.18	**0.91 ± 0.15**	**0.10 ± 0.28**	1.42 ± 2.60
Proposed	(25,50)	**2.88 ± 3.78**	**0.55 ± 0.71**	**0.94 ± 0.08**	**0.10 ± 0.16**	1.61 ± 1.92
Proposed	(25,100)	**1.57 ± 1.92**	0.43 ± 0.64	**0.93 ± 0.10**	**0.08 ± 0.14**	0.89 ± 1.07
Proposed	(50,25)	2.51 ± 4.64	0.31 ± 0.44	**0.88 ± 0.20**	0.09 ± 0.18	1.42 ± 2.64
Proposed	(50,50)	**2.42 ± 4.62**	**0.24 ± 0.34**	**0.93 ± 0.12**	**0.10 ± 0.27**	1.35 ± 2.58
Proposed	(50,100)	**2.98 ± 4.75**	**0.35 ± 0.54**	**0.95 ± 0.06**	**0.13 ± 0.34**	1.58 ± 2.52
Proposed	(100,25)	**2.06 ± 3.31**	0.22 ± 0.18	**0.94 ± 0.12**	**0.06 ± 0.16**	1.25 ± 2.08
Proposed	(100,50)	**2.41 ± 4.24**	0.23 ± 0.27	**0.96 ± 0.05**	**0.07 ± 0.16**	1.35 ± 2.43
Proposed	(100,100)	**2.31 ± 3.60**	0.24 ± 0.28	**0.96 ± 0.06**	**0.08 ± 0.16**	1.32 ± 2.10
Proposed-CF	(25,25)	3.55 ± 6.51	**0.49 ± 1.05**	0.79 ± 0.19	0.37 ± 1.28	1.23 ± 1.94
Proposed-CF	(25,50)	3.89 ± 4.82	0.81 ± 1.21	0.85 ± 0.09	0.32 ± 0.56	1.75 ± 2.15
Proposed-CF	(25,100)	2.47 ± 2.74	0.59 ± 0.92	0.75 ± 0.21	0.17 ± 0.26	1.13 ± 1.34
Proposed-CF	(50,25)	3.12 ± 4.76	0.63 ± 1.03	0.78 ± 0.21	0.19 ± 0.30	1.49 ± 2.52
Proposed-CF	(50,50)	2.92 ± 4.64	0.40 ± 0.90	0.85 ± 0.16	0.16 ± 0.35	1.33 ± 2.39
Proposed-CF	(50,100)	3.38 ± 4.87	0.43 ± 0.70	0.87 ± 0.14	0.15 ± 0.26	1.57 ± 2.45
Proposed-CF	(100,25)	2.32 ± 2.99	0.29 ± 0.22	0.83 ± 0.24	0.08 ± 0.16	1.23 ± 1.89
Proposed-CF	(100,50)	2.62 ± 4.34	0.27 ± 0.38	0.88 ± 0.14	0.10 ± 0.20	1.38 ± 2.55
Proposed-CF	(100,100)	2.57 ± 3.75	0.34 ± 0.60	0.90 ± 0.11	0.09 ± 0.19	1.37 ± 2.28

TABLE XXIV

IHDP DISCONNECTED REGIME ($m_1 = 0$): FULL METRICS.

Method	Budget	PEHE	ATE err	Spearman	Regret	ECE
ProxyOnly	25	2.64 ± 2.65	1.06 ± 1.61	0.65 ± 0.25	0.16 ± 0.27	1.86 ± 1.75
ProxyOnly	50	3.46 ± 4.88	1.59 ± 3.11	0.74 ± 0.18	0.16 ± 0.27	2.25 ± 3.23
ProxyOnly	100	3.50 ± 4.27	1.21 ± 1.70	0.77 ± 0.22	0.19 ± 0.28	1.80 ± 1.92
ProxyOnly	200	3.79 ± 4.53	2.05 ± 3.16	0.81 ± 0.18	0.22 ± 0.35	2.41 ± 3.09
IPW-Transport	25	2.30 ± 3.08	0.81 ± 1.22	0.79 ± 0.20	0.05 ± 0.10	1.43 ± 1.69
IPW-Transport	50	3.36 ± 5.84	1.38 ± 2.60	0.81 ± 0.20	0.16 ± 0.59	2.08 ± 3.58
IPW-Transport	100	4.87 ± 6.95	1.91 ± 3.63	0.76 ± 0.27	0.38 ± 0.87	2.91 ± 4.38
IPW-Transport	200	4.79 ± 6.95	1.95 ± 4.10	0.83 ± 0.18	0.35 ± 0.96	3.01 ± 4.72
EntropyBal	25	2.82 ± 3.02	1.19 ± 1.43	0.72 ± 0.22	0.13 ± 0.25	1.83 ± 1.59
EntropyBal	50	3.61 ± 5.65	1.58 ± 2.64	0.76 ± 0.23	0.24 ± 0.81	2.25 ± 3.29
EntropyBal	100	5.47 ± 7.99	2.30 ± 4.33	0.75 ± 0.26	0.59 ± 1.57	3.31 ± 5.08
EntropyBal	200	5.63 ± 8.62	2.17 ± 4.92	0.74 ± 0.29	0.57 ± 1.69	3.57 ± 6.16
OM-Transport	25	2.28 ± 3.41	0.87 ± 1.42	0.83 ± 0.16	**0.04 ± 0.08**	1.43 ± 1.81
OM-Transport	50	3.37 ± 6.43	1.26 ± 2.76	0.84 ± 0.20	0.15 ± 0.49	2.10 ± 4.08
OM-Transport	100	4.18 ± 5.96	1.46 ± 2.28	0.83 ± 0.24	0.21 ± 0.49	2.46 ± 3.36
OM-Transport	200	4.50 ± 6.64	1.53 ± 3.08	0.89 ± 0.12	0.27 ± 0.70	2.87 ± 4.46
Proposed-B	25	**2.11 ± 3.44**	**0.74 ± 1.31**	**0.88 ± 0.14**	**0.04 ± 0.07**	**1.29 ± 1.85**
Proposed-B	50	**3.19 ± 6.23**	**1.15 ± 2.69**	**0.91 ± 0.14**	0.18 ± 0.57	**2.00 ± 4.04**
Proposed-B	100	**3.90 ± 5.88**	**1.09 ± 1.77**	**0.88 ± 0.18**	**0.17 ± 0.27**	**2.14 ± 3.04**
Proposed-B	200	**4.33 ± 6.59**	**1.45 ± 2.92**	**0.91 ± 0.12**	**0.26 ± 0.61**	**2.75 ± 4.48**